MIL-M-38310B (USAF)
1 July 1971
SUPERSEDING
MIL-M-38310A (USAF)
15 July 1966

S

MILITARY SPECIFICATION

MASS PROPERTIES CONTROL REQUIREMENTS FOR MISSILE AND SPACE VEHICLES

1. SCOPE

1.1 General. - This document provides a means to establish a system for the management of mass properties during the conceptual, validation, development, production and deployment of surface-to-surface ballistic vehicles and space vehicles, or portions thereof. It is designed to permit the acquisition of systematized, traceable, verifiable and controllable mass properties of vehicle systems, to facilitate rapid establishment and reporting of inputs for the weight/performance relationship, and to enable parametric extrapolation from the reported systems to newly evolving systems.

1.2 Limitations. - The elements of this specification are not to be placed on contract in toto, but are listed to provide a comprehensive collection of elements from which selection may be made as appropriate for the discrete requirements of each program.

2. APPLICABLE DOCUMENTS

2.1 Effective Issue. - The following document of the issue in effect on the date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

Military
MIL-P-116 Preservation, Methods of

3. REQUIREMENTS

3.1 Objectives

3.1.1 General. - A system of mass properties management shall be established to ensure fulfillment of the program mass properties objectives and compliance with contractual requirements.

3.1.2 Development of Objectives. - The contractor shall develop achievable mass properties objectives and shall assist the procuring activity in specifying General System Specification mass properties requirements, and their proper allocation to the Configuration Element(s) Specification(s) requirements. During the initial development of, or any subsequent change to, mass properties objectives, the contractor shall give particular attention to substantiating data, mass properties dependent design information and the critical mass properties uncertainties analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.1.3 Development of Limits. - Throughout the contract life cycle, the contractor shall develop and maintain cognizance of the vehicle mass properties limits whether established by contract, system performance or by subsystem design. The compatibility of total system mass properties requirements with the configuration element(s) requirements shall be assessed to assure that the program objectives are met.

3.2 Studies, Planning, Cognizance and Control

3.2.1 Studies

3.2.1.1 Trade Studies Methodology. - Adequate methodology and tools shall be developed, or existing methodology and tools adapted, to support trade studies involving mass properties inputs. Such methodology and tools may be curves, computer programs, or any suitable means for relating major design parameters to significant mass properties and providing parametric assessment of inter-system and intra-system mass properties during trade studies leading to item selection.

3.2.2 Planning

3.2.2.1 Program Plan. - The contractor shall develop a master plan for a system of mass properties control which will ensure fulfillment of the program mass properties objectives. Both technical and organizational needs shall be considered. Necessary supporting plans shall be identified.

3.2.2.2 Supporting Plans. - The contractor shall develop supporting plans for the system of mass properties control to ensure the effective execution of control functions such as analysis, verification, improvement, reporting and operational field support. The impact of changes to critical mass properties on these plans shall be assessed and changes to the plans shall be formulated.

3.2.2.3 Plan Review. - The contractor shall be prepared to review any plan concerning the program system of mass properties control with the procuring activity.

3.2.2.4 Coordination. - After go-ahead and prior to first report submittal, the contractor shall be prepared to review with the procuring activity:

(a) The interpretation of contractual requirements.

(b) The planned mass properties management program.

3.2.2.5 Personnel. - Qualified personnel shall be assigned the responsibility and authority to assure the establishment and maintenance of mass properties objectives, and the effective planning and execution of mass properties control functions.

3.2.2.6 Planned Statement of Work. - A proposed Statement of Work for the subsequent phase of the program shall be developed by the contractor. It shall identify the tasks necessary for a system of mass properties management that will be in consonance with the program objectives of the particular phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.2.2.7 Growth Depletion. - The contractor shall develop a planned approach to the depletion of the growth values utilized in the derivation of the System Mass Properties Base for Control and Reporting. It will identify the growth remaining at significant points in the contract life cycle.

3.2.3 Cognizance

3.2.3.1 Monitoring. - Each item, and the operating characteristics of transfer, loading, and utilization systems shall be analyzed and monitored continuously to establish current mass properties and their trend.

3.2.3.2 Uncertainty Analyses. - The uncertainty associated with each of the Critical Mass Properties shall be established by analytical methods. These established uncertainties shall be used in dispersion studies, for analytical verification of computed nominal values, and in the identification of system elements which warrant verification by experimental methods.

3.2.3.3 Subcontractor Surveillance. - The contractor shall be responsible for the adequacy of each subcontractor's system of mass properties control. An appropriate mass properties section, comparable to and compatible with the contractual requirements, shall be incorporated in each procurement document affecting system mass properties.

3.2.3.4 Associate Design Activity and GFE Suppliers Interfaces. - Associate Design Activities and GFE Suppliers shall be responsible for the interchange of sufficient mass properties data to support the integration of sub-unit mass properties into the complete unit mass properties and shall promptly respond to requests from the interfacing and integrating contractors for information required by them in satisfaction of contractual requirements.

3.2.3.5 Supplemental Data. - The contractor shall be prepared to provide, in an expeditious manner, existing supplemental data or informal inputs for any report.

3.2.3.6 Trade Studies Inputs. - The contractor shall determine and maintain, in a form available for review by the procuring activity, the following:

(a) System mass properties inputs used in trade studies and other screening processes for including or excluding designs for future study.

(b) The effect on system mass properties of each trade study, whether or not performed as a mass properties trade.

3.2.3.7 Identification of Changes. - The contractor shall provide and maintain in a form available for review by the procuring activity, an account which identifies the nature, magnitude and sequence of each change in proposed mass properties from the initiation to the termination of the program.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

3.2.3.8 Review Meeting Support. - Technical support shall be provided for review meetings involving system mass properties, viz:

(a) In-house management review meetings.

(b) Associate contractor and/or sub-contractor interface meetings.

(c) Review meetings with the procuring activity.

3.2.3.9 Relationships. - Mass properties data provided by the contractor in response to contractual requirements shall employ the definitions of paragraph 6.3 and the functional code of Appendix B, and shall be presented in a manner which will facilitate the rapid establishment of their relationship to current performance, design and mission functions of the item.

3.2.3.10 Interface with Associate and Integrating Contractors. - The contractor shall be responsible for the timely interchange of sufficient mass properties information to support the integration of subsystem mass properties into system level mass properties by requesting needed information from the pertinent contractor(s) and agency(ies) and by responding promptly to like requests for information from the contractor(s) and agency(ies).

3.2.3.11 System Engineering. - All Associate Contractors reported mass properties shall be evaluated by the contractor and integrated to provide management visibility of system level mass properties status, of major problems encountered, and/or progress made in the mass properties management program. The contractor's evaluation of reported system mass properties shall ensure the technical accuracy of these inputs for integration purposes and their compatibility with current program baselines and objectives. When the interests of program data currency and compatibility require adjustments to these inputs prior to system level integration such adjustments shall be made and the affected supplier of the data shall be notified promptly of the change, the reason for the change, and the proposed manner of incorporation. Every effort shall be made to obtain approval from the affected supplier for such adjustment prior to incorporation, however approval must be obtained in all cases.

3.2.4 Control

3.2.4.1 System Mass Properties Base

3.2.4.1.1 Specified Base. - The system mass properties base shall be as specified by the procuring activity.

3.2.4.1.2 Specification Base. - The system mass properties base shall be the system mass properties as defined by the approved system specification.

3.2.4.1.3 Reference Base. - The system mass properties base at the initiation of the contract shall be defined for reference purposes only. This base has no control implications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.2.4.2 Improvement Action

3.2.4.2.1 Means to Meet Specified Mass Properties. - Mass properties control personnel, working with analytical, design and management personnel, and considering such constraints as reliability, performance, schedule and cost, shall determine means to insure that specified mass properties are met or bettered.

3.2.4.2.2 Improvement Potentials. - A summary of all improvement potentials which could be used to offset mass properties or performance degradation shall be established and maintained. Mass properties analyses for all alternative design configurations shall be conducted.

3.2.4.2.3 Problem Definition. - The procuring activity shall be notified immediately of any problem requiring improvement action, with a statement of the problem and the potential effect on mass properties and performance when:

(a) Program objectives are endangered.

(b) Established control parameters have been reached or exceeded.

3.2.4.2.4 Corrective Action. - The contractor shall take, or recommend to the procuring activity, actions which will correct the recognized deficiencies.

3.2.4.2.5 Limiting Condition Analyses. - Analyses shall be performed to assist the procuring activity in the establishment and maintenance of contractual mass properties limits. Analyses shall be performed to establish and maintain other mass properties limits created by mandatory and acceptable system objectives and functions. Analyses shall consider current system design and operating characteristics.

3.2.4.2.6 Document Release. - Documents controlling the manufacture and procurement of system components shall be approved, prior to release, by personnel responsible for the contractor's mass properties control effort. Such approval shall signify component acceptability in satisfaction of system mass properties objectives and the currency of the mass properties accounting system. Release without approval shall be so noted.

3.3 Verification

3.3.1 Verification of Critical Mass Properties. - Each Critical Mass Property and its conformance to limits shall be verified by the contractor. Verification shall be performed by approved analytical or experimental methods, or by a combination of both.

3.3.2 Contract End Item Verification. - Contract End Item mass properties shall be verified by approved analytical or experimental methods, or by a combination of both.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.3.3 Detail Part Verification. - Detail part mass properties shall be verified by approved analytical or experimental methods, or by a combination of both.

3.3.4 Measurement Conditions. - Item measurements performed in accordance with procuring activity approved procedures shall satisfy the following conditions at the time of measurement:

(a) The item shall simulate the dry flight condition and be at least 95 percent complete by weight, excluding hazardous components or components not normally installed at the measurement site.

(b) The manufacturing status of the item shall be documented.

(c) A representative of the contractor's mass properties control section shall be present.

3.3.5 Notification of Measurement. - In all cases of measurements to be performed in accordance with procuring activity approved procedures, the contractor shall notify the procuring activity of the time and place at least one week in advance.

3.3.6 Data Records. - Mass properties verification data shall be recorded and maintained in a form available for review by the procuring activity. Actual mass properties data records shall be provided, in accordance with approved procedures, for each major measurement performed.

3.3.7 Post-Flight Analyses. - The contractor shall determine, by analysis of post-flight data, the actual system mass properties for the initial and terminal conditions of flight and compare these actual conditions to the planned conditions. Each difference in mass properties shall be itemized, explained and the effect on system performance capability defined. Adequate instrumentation and procedures shall be provided, subject to approval by the procuring activity, to ensure the capability for post-flight determination of system mass properties.

3.3.8 Field Support of Launch Operations. - The contractor shall provide adequate mass properties support to launch operations. This support shall maintain mass properties inputs to flight planning which are in agreement with the actual vehicle configuration and the planned loading and utilization of fluids.

3.4 Documentation. - Reports shall be submitted in accordance with this document, unless otherwise specified by the procuring activity.

3.4.1 Types of Reports Required. - The required reports are described by the following general types. Their specific composition is set forth in Section 3.4.2 and may vary according to the phase of procurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.1.1 Detail Mass Properties Reports. - Reports in this category provide comprehensive detail reference documentation of the mass properties as they exist at initial, intermediate and final stages of procurement. Submittals occur at specified times in the Conceptual, Validation, Development and Production phases. Vehicles or vehicle blocks reported upon shall be those specified by the procuring activity.

3.4.1.2 Mass Properties Status Reports. - Reports in this category provide a current status of the mass properties, of changes since the last report, and of problems encountered and progress made in tasks associated with the mass properties control program. Submittals occur at specified intervals during the Conceptual, Validation, Development, Production and Deployment phases. Submittals also occur at specified events during flight preparation and launch of each vehicle. Vehicles or vehicle blocks reported upon shall be those specified by the procuring activity.

3.4.1.3 Procedural Reports. - Reports in this category provide outlines of the contractor's approach for satisfying the requirements of this document. Initial submittals occur sufficiently in advance of implementation to permit their review and approval.

3.4.1.4 Miscellaneous Reports. - Submittal dates for the following reports in this category cannot be specified in this document.

3.4.1.4.1 Contract Change Proposals. - Information to evaluate and substantiate the effect on vehicle mass properties from proposed changes. Submittal occurs with the Change Proposal.

3.4.1.4.2 Test Results. - Results of testing to verify parameters for the mass properties of fluids or propellants submitted in accordance with the design activity's Mass Properties Verification Plan.

3.4.1.4.3 Operational Targeting Data. - Mass Properties targeting information submitted in accordance with the Operational Support Plan for Targeting.

3.4.2 Report Format

3.4.2.1 General. - This section establishes the basic format requirements for reports.

3.4.2.1.1 Unit System of Measure. - Data contained in the required mass properties reports shall be in the unit system specified by the procuring activity.

3.4.2.1.2 Forms. - Report elements which require the use of forms shall employ forms similar to those illustrated in Appendix A.

3.4.2.1.3 Coding. - To provide a uniform basis for mass properties comparison to facilitate the preparation of mass properties summaries, the contractor shall employ the functional code and nomenclature in accordance with paragraph 6.3. Where contract responsibility includes items comprising more than one stage, the components for each stage shall be tabulated separately in each report.

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.2.1.4 Coordinate Axes. - To provide a common reference basis for all mass properties, performance and design data, the coordinate axes and notation system shown in Figure 1 shall be used. The X axis shall always be the thrust axis of the vehicle when on the launch pad. Insofar as it is practical, all spacecraft shall use coordinate axes consistent with their position on the launch vehicle on the pad. The vertical and lateral reference datums shall be the planes defined by the intersection of the X-Y axes and the X-Z axes, respectively.

3.4.2.2 Report Elements. - Unless otherwise specified by the procuring activity, reports shall be composed of those applicable elements listed in Table 1, Page 25, and the element shall be as described in Section 3.4.3.

3.4.3 Report Element Description

3.4.3.1 Title Page. - A title page shall contain the following information, as applicable:

(a) Report number.

(b) Type of submittal.

(c) Vehicle flight number.

(d) Step, stage or module.

(e) Applicable serial numbers.

(f) Date of issuance.

(g) Actual date of data reported.

(h) Contractor's name.

(i) Mission identification.

3.4.3.2 Table of Contents. - A Table of Contents is a listing of the report elements and their location within the report.

3.4.3.3 Review for Management. - The Review for Management shall present a summary of the significant material contained in the report.

3.4.3.3.1 Management Review Tabulations. - Reports containing a Mass Properties Summary shall include a concise management review tabulation presenting the following, as applicable:

(a) Weights - revised Specified Weight Base, last current status and current status, together with the change from last to current for:

(1) Total dry weight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

(2) Residuals and reserves.

(3) Dry weight items jettisoned and ablated during flight.

(4) Subtotal of (1) less (3) (minimum burn-out weight).

(5) Total available design thrust propellants.

(6) Ignition weight.

(7) Maximum loadable tank capacity (main propellants).

(b) Control Parameter Encroachment - for each current mass property which has reached or exceeded its control parameter, specify the following:

(1) Control parameter and its associated limit.

(2) Current nominal mass property value.

(3) Accuracy or uncertainty of the current nominal value.

(c) Highlights of newly encountered problems.

(d) Graphs, beginning at the proposal stage, presenting the historical record for dry weight, minimum burn-out weight, launch weight and contractor responsibility and procuring activity responsibility weight changes, as a function of time.

3.4.3.4 Organization. - Furnish organization charts and descriptive material to identify the key personnel responsible for mass properties control, their relationship to other elements of the organization and the percentage of their time devoted to the contract. These data shall be up-dated whenever changes occur.

3.4.3.5 Mass Properties Summary. - Summarize current mass properties detail data in an orderly build-up to the dry, terminal, lift-off, ignition, and gross weight conditions, as applicable. Expenditures, reserves and residuals shall be clearly identified in the build-up. This information shall be reported on a form similar to Form 1, Part I (Appendix A) for Items (a) through (j) and (m) below and on a form similar to Form 1, Part II (Appendix A) for items (a), (b), (f), (k) and (l) below. The minimum acceptable amount of information reported shall be as follows:

(a) Code - first and second generation functional code, except for propulsion systems which shall be carried to at least the third generation.

(b) Description - appropriate functional code nomenclature or summary title.

(c) Specified Weight Base - original specified weight base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

(d) Procuring Activity and Government Furnished Equipment (GFE). Out of scope changes resulting from procuring activity initiated changes and/or GFE changes.

(e) Revised Specified Weight Base - the sum of (c) and (d).

(f) Current Weight.

(g) Changes, last to current, Contractor Responsibility in-scope changes in current weight since last report.

(h) Changes, last to current, procuring activity responsibility out of scope changes in current weight since last report.

(i) Percentage Breakdown of Current Weight - Percent of reported current weight which was based on estimates, layout analyses, drawing calculations, or actual weights.

(j) Note Number - Reference numbers for explanatory notes, remarks, or change identification.

(k) Center of Gravity - current centers of gravity.

(l) Moment of Inertia - current moments of inertia.

(m) The last two line entries on Form 1, Part I (Appendix A), shall summarize the contractor responsibility and procuring activity responsibility portions of total weight for items (c), (e) and (f).

3.4.3.6 Detail Mass Properties. - Provide a complete breakdown of the item's current mass properties. These data shall be prepared on a form similar to Form 2, Parts I and II (Appendix A), and shall include, but not be limited to, the following:

(a) Code - applicable first, second and third generation functional code, except for propulsion systems which shall be carried to at least the fourth generation.

(b) Description - applicable functional code and summary titles.

(c) Class - third generation weight as estimated, calculated or actual.

(d) Current Mass Properties.

3.4.3.7 Change Analysis and Improvement Potentials

3.4.3.7.1 Mass Properties Change Analysis. - An explanation of significant changes in current mass properties data and of all changes in the Specified Weight Base shall be reported on a form similar to Form 3, Parts I and II (Appendix A). All changes in Government Furnished Equipment shall be identified and explained. Each

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

change shall be cross-referenced to the Mass Properties Summary Report Element (Paragraph 3.4.3.5) and the Sequenced Mass Properties Data Element (Paragraph 3.4.3.10) as applicable. The minimum acceptable amount of information shown in this element shall be as follows:

(a) Note Number - reference number corresponding to the note number in the Mass Properties Summary Element or the Sequenced Mass Properties Data Element.

(b) Code - all functional code numbers affected.

(c) Change, Total - total change in mass properties for each functional code.

(d) Effective Point - vehicle effectivity point, as applicable.

(e) Change, Contractor Responsibility - change in mass properties as a result of in-scope changes.

(f) Change, Procuring Activity Responsibility - change in mass properties resulting from out of scope changes.

(g) Remarks - Description of each change, the authority for the change and the effects of the change on the mass properties dependent design information. Changes that result from reconciliation of predicted values with actual measurement of configuration elements, or major portions thereof, shall be fully identified.

3.7.3.7.2 Pending Mass Properties Change Analysis. - Each significant pending change in mass properties, including contract change proposals, shall be reported on a form similar to Form 3, Parts I and II (Appendix A). Separate forms will be utilized to report changes established subsequent to the basic report close-out date, corrective action changes recommended by the contractor to correct recognized deficiencies and any other pending change. The following minimum information shall be submitted with each pending change.

(a) Change Identification - case number assigned by the contractor or the customer.

(b) Code - all applicable first and second generation code numbers affected.

(c) Change - current and pending mass properties for each affected code together with the total mass properties change.

(d) Effective Point - vehicle number or approximate date when pending change will be incorporated.

(e) Remarks - explanation of each change, including reference correspondence, change order number, drawing number and other pertinent data.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.3.7.3 Summary of Improvement Potentials. - Tabulate all mass property improvement potentials which could be used to offset degradations. Include Identification or Case Number and best estimates of effects on mass properties, schedules, costs, reliability and effectivity. Indicate whether technical feasibility is established or requires further study. Items which have been rejected shall be accumulated separately, together with the reasons for rejection.

3.4.3.8 Unresolved Problems. - All problems that may affect mass properties shall be reported at the earliest possible time including the system affected, action being taken and an estimate of when the problem will be resolved.

3.4.3.9 Historical Log. - Provide a record of the item's mass properties configuration at the time of the first major measurement and for each subsequent change on a log form similar to Form 4, Part I and Part II (Appendix A). The first log entry shall record the "as measured condition" mass properties of the item and identify the Actual Mass Properties Data Record (3.4.3.18) from which the data were extracted. A copy of the shortage list contained in the variance items (3.4.3.18.2c) shall accompany each copy of the log from the time of the first entry. Subsequent log entries shall be recorded each time a part is added, removed or relocated, with corresponding adjustments made in the shortage lists, when applicable. If the procuring activity has not established measurement requirements for center of gravity and inertia data, calculated data may be used. Log sheets shall be sequentially numbered. Log entries shall be in chronological order.

3.4.3.10 Sequenced Data. - This report element shall contain weight, balance and inertia data on the step/stage/vehicle for each major event of the mission and shall be reported on a form similar to Form 5 (Appendix A). The contractor shall use current detail data and engineering information to develop mass properties for each mission event. These data shall start with the step/stage/vehicle at ground ignition and continue throughout the mission to separation, impact and recovery, as applicable. The mass properties and code numbers of items subtracted from the totals in getting from one event to another shall be shown. The information reported in this section shall be the following, as applicable:

(a) Code - functional code number.

(b) Description - functional code title or mission event.

(c) Weight - current weight. Significant changes since the previous report shall be underlined and cross-referenced to the change analysis element (Paragraph 3.4.3.7).

(d) Center of Gravity - center of gravity location measured from the reference datum. Significant changes since the previous report shall be underlined and cross-referenced to the change analysis element.

(e) Moment of Inertia - current moment of inertia. Significant changes since the previous report shall be underlined and cross-referenced to the change analysis element.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

(f) Product of Inertia - where required, columns may be added for the presentation of products of inertia.

(g) Angles of Intersection - where required, between the principal and geometric axes.

3.4.3.11 Powered Flight Data. - This report element shall contain mass properties data used to derive performance information. It shall include, but not be limited to, the following:

(a) Tabulation or plot of weight, center of gravity, pitch, yaw and roll moments of inertia versus mission time.

(b) Assumptions made and data used to derive these mass properties (propellant weights, propellant duration, liquid inertias during flight, engine parameters, ejection time on all items, stage burning time).

(c) Reference shall be made as applicable to other report elements from which the above data may be extracted or based upon.

3.4.3.12 Diagrams. - Include sketches, curves and drawings (elevation and plan) sufficient to supply the following minimum information:

(a) Principal structural stations and structural interface locations.

(b) Reference datum for the 3-axes moment arms, showing quadrant numbers and gimbal plane stations.

(c) Significant lengths, heights, widths and diameters.

(d) Sections of the vehicle as sectionalized for shipping and handling.

(e) Locations of jig, leveling and weighing points.

(f) Thrust reaction stations, giving direction of thrust and points of application. Where the thrust direction is not fixed the maximum angle of variation shall be specified.

(g) Center of gravity envelope for stages and the complete vehicle, as applicable.

3.4.3.13 Substantiating Data. - Validate the reported mass properties values, including contingency and growth, by including the analytical, statistical or empirical methods used in their derivations. When the complexity of the equations and computer programs prevents inclusion in the document, a summary shall be presented and the detail methods shall be readily available for review by the procuring activity. Where values are determined by an accumulation of separate equipment groups, values shall be given for each component. These data shall be updated whenever changes occur.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.3.14 Government Furnished Equipment. - A separate tabulation of all GFE shall be prepared for each vehicle stage or module of each flight number showing the drawing number and description, serial number where applicable, number required, and the unit, total dry and total wet specified mass properties. Provisions shall be included to record the actual unit weight and total weight, and the difference between total specified and total actual weights. Changes to the specified weights of GFE items shall be listed.

3.4.3.15 Computer Cards or Tapes. - These are detail mass properties to be prepared in accordance with the procuring activity's instructions.

3.4.3.16 Mass Distribution. - Data shall be prepared for all critical conditions which depend on mass distribution for their analyses and shall be in tabular or graphical form presenting the following information, including supporting calculations:

(a) Weight per unit of longitudinal length representing distributed loads.

(b) Concentrated loads and related reaction points.

(c) Cantilevered loads and related reaction points, moments and moment direction (plus or minus).

3.4.3.17 Design Data for Mass Properties Analysis

3.4.3.17.1 Mass Properties Dependent Design Information. - These data include major dimensional factors and design criteria used in the development of the reported mass properties. Appendix C presents a list which shall be used as a guide for data submitted.

3.4.3.17.2 Structural Increments for Design Features. - Net structural increments for special functions, installations and other design features considered to be other than basic on any major design group shall be included. Insofar as practicable, the amount buried or masked in "basic structure" and quantitatively evaluated by some rational procedure should be listed for such items as body joints, major cutouts, removable structural panels, etc. Values shall be accompanied by amplifying and clarifying remarks or data verifying the means of determination.

3.4.3.18 Actual Data Records

3.4.3.18.1 General. - Records for each major mass properties measurement performed in accordance with the procedures approved by the procuring activity shall include:

(a) Location where measurements were performed, signature of authorized individual responsible for the entries, date and time of entries, date of last equipment calibration, document number of the approved measuring procedures and identification of the item measured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

(b) Provision for the signature of a procuring activity witness on each page of the data record that includes measured data.

3.4.3.18.2 Records for Dry Mass Properties Measurements. - Records for dry mass properties measurements shall include:

(a) Tables showing the applicable scale readings: tare; net weight; moment arm: moment for longitudinal, vertical and lateral center of gravity; and calculations showing the derivation of the "as measured" weight and center of gravity condition from the measurements.

(b) Measurements taken for the determination of moment and product of inertia and calculations showing the derivation of the "as measured" inertias from the measurements.

(c) Variance Items - A list of items, including a Shortage List, for the weight, center of gravity and inertia data to be added to or subtracted from the "as measured" condition to obtain the actual "dry weight" determination for launch condition.

(d) Diagrams of measuring equipment and related fixtures showing pertinent dimensions and other data required for the determination of the "as measured" weight, center of gravity, and moment and product of inertia.

3.4.3.18.3 Records for Fluid Mass Properties Measurements. - Records for fluid mass properties measurements shall include:

(a) Explanation of deviations from test plan.

(b) Summary of data.

(c) Evaluation of results.

(d) Final conclusion of results.

(e) Recommendations.

3.4.3.19 Summary of Critical Mass Properties. - Provide a table stating the relationship between the current value and its limiting value for all of the Critical Mass Properties for each mission, including at least the following:

(a) Limiting mass property value.

(b) Current mass property value.

(c) Accuracy or uncertainty of the current value.

(d) Existing tolerance on the current value.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

(e) Reference to the current Critical Mass Property Uncertainties analysis and to the current Mass Properties Limits analysis.

3.4.3.20 Evaluation of Flight. - Compare the actual launch mass properties and the actual mass properties variations throughout the flight with the predicted mass properties flight data. In liquid propellant systems the evaluation shall include analytical extrapolation from the actual propellants at thrust termination to the outage condition which would have existed had burning been to depletion. An evaluation of the critical mass properties, uncertainties and other pertinent data shall be included in the final post flight reports to substantiate mass properties contained in the contractor's Final Flight Evaluation Report. Quick Look reports should be confined to gross information and analysis based on the best available sources to confirm either normal flight mass properties conditions or apparent anomalies.

3.4.3.21 Uncertainties. - Provide sufficient data for, and analysis of the critical mass property uncertainties to be used in association with studies of trajectory dispersions and other limiting conditions. The information is to be organized to permit determination of those sources of error which are large enough to warrant test verification of nominals and uncertainties and to serve as a guide in the approach to any testing required. This analysis shall include at least the following for each mission:

(a) Uncertainties in the critical mass property conditions of the system shall be summarized and substantiated by detailed analysis of each contributing error source. Each quantity for which uncertainties are summarized shall be broken down to a level corresponding with its principal elements.

(b) Analysis or discussion to describe the derivation of each measured, calculated or otherwise assigned uncertainty along with the nominal value assumed for the analysis. When included in the nominals being subjected to analyses weight growth and contingency allowances shall be identified.

(c) Error sources selected shall be described and shall include uncertainties anticipated or indicated by analysis of the following, as applicable:

(1) Nominal inert and fluid weights together with the launch procedures, ambient conditions, in-flight changes and manufacturing variations affecting same.

(2) Instrumentation performance associated with verification or testing of the above. All uncertainties stated as errors or dispersions shall be accompanied by statements of the confidence of achieving the particular level of error. If the distribution of the uncertainties is known to be a normal distribution, then the confidence may be stated in terms of a multiple of the standard deviation. Otherwise, the probability of not exceeding the particular error value must be stated.

3.4.3.22 Limits. - Mass Properties Limits shall include the following:

(a) A tabulation of the limiting conditions and the associated mass properties limits which are established by contract. Include reference to the contract document and page where the requirement is placed.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

(b) A tabulation of the limiting conditions and the associated mass properties limits which are established by mandatory and acceptable system objectives and functions under the current design and operating characteristics.

(c) Analyses supporting and establishing the allowable limiting values of each mass property condition tabulated in (b) above.

3.4.3.23 Fluid and Propellant Loads. - For each fluid and propellant load reported, present an inventory of the load and the maximum loaded weight together with a statement of the parameters and other critieria employed in their derivation, sufficient to substantiate the reported values. A guide for presenting the required data is shown in Appendix D.

3.4.3.24 Capacity and Loading Information. - For each fluid and propellant, provide system capacity data, loading and associated information required to predict the loaded, expended and unexpended values for mission planning. A guide for presenting the required data is shown in Appendix F.

3.4.3.25 Verification Plan. - Provide the plan for contractor verification of all system mass properties which are critical. The plan shall include the following:

(a) Identification of the system mass properties, and elements thereof, for which experimental verification is, or is not, proposed. Justification shall be provided for either case. Test plans for proposed experimental measurements shall be provided, incorporating an error analysis of the equipment and procedures to be used. Appendix E is a guide for preparing the required information.

(b) Description of the planned procedures to determine, record and report all post-measurement mass property changes resulting from changes in configuration and in operating conditions affecting the loading and utilization of propellants and fluids. Description of planned communication methods to assure rapid pre- and post-flight reporting of these changes to the procuring activity and the contractor's principal points of contact in the field and at the home plant shall be indicated.

(c) Description of the planned procedures to assure that inflight mass property changes are promptly detected and determined during flight and that any decision for inflight change to mission performance objectives is based on correct mass properties. This part of the plan shall include an examination of contingencies which may logically cause significant deviations in system mass characteristics, provisions for improving mass characteristics by emergency jettison or load transfer, and provisions for rapidly displaying results to the crew or controlling station.

(d) Proposed procedures for the identification and maintenance of valid performance mass property inputs to weapon system targeting parameters. These procedures shall be conceived to be implemented by the procuring activity of the appropriate Engineering Service Agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.3.26 Operational Targeting Data. - Submit mass properties targeting data in accordance with the proposal of 3.4.3.25(d). Present the actual and statistical mass properties data which are required to support the operational targeting concept. Accompanying these data shall be the maintenance procedures and specifications to insure its upkeep by the appropriate Service Engineering Agency.

3.4.3.27 Proposed Statement of Work. - Present a proposed Statement of Work for the subsequent contract phase. Task statements shall, where possible, be derived from and be numerically identified with the appropriate paragraphs of this specification. New or different tasks shall also be included. A proposed Contract Data Requirements List constitutes part of the Statement of Work. In the case of an integrating contractor:

(a) Present a CDRL of the Integrating Contractor's requirements for data from Associate Contractors and procuring activity needed to enable the Integrating Contractor to perform his mass properties integrating role.

(b) Describe, in contractual language, the supporting tasks required of the Associate Contractors and procuring activity needed to enable the Integrating Contractor to perform his mass properties integrating role.

3.4.3.28 Program Plan. - Present a master plan of a system of tasks that describes how the requirements of the specific Statement of Work will be implemented. The plan will consider all mass properties tasks, either implied or stated, to the lowest level of paragraph within the Statement of Work and shall respond by either:

(a) Stating how the task will be performed.

(b) Referencing the appropriate paragraph within the plan where description of the implementation function is made. This may be done where operations are repetitive or where centralization of all mass properties functions makes for a more cohesive and comprehensive plan. The paragraph numbering and content shall agree with that of the Statement of Work. Conformance to, or deviations from, the Contract Data Requirements List shall be stated together with the rationale for any desired variance.

3.4.3.29 Trade Study Data. - Trade study data shall include the following where applicable:

(a) A list of the effect on mass properties of each trade study whether or not performed directly as a mass properties trade.

(b) Identification of each listed trade study by title.

(c) Description of the methodology and tools employed in deriving the results of the trade.

(d) Any mass properties criteria used in system screening or selection.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.3.30 Growth Allowance Depletion. - Present, either graphically or in tabular form, the proposed approach to depletion of the total growth allowance. The growth remaining at significant points in the contract life cycle shall be identified.

3.4.3.31 Data Sources. - All reports shall include a listing of the pertinent references, such as data sources, reports, correspondence, substantiating documents and any other material germane to the report.

3.4.4 Report Submittal Requirements

3.4.4.1 Schedule of Submittals. - Unless otherwise specified by the contractor, Table I constitutes the required submittal schedules for reports. The schedules reflect required dates for receipt of final contractor approved documents at the office of the procuring activity where technical management of the program resides.

3.4.4.2 Distribution. - Unless otherwise directed by the procuring activity distribution is required to the organization which has technical mass property cognizance of the program and to such other distribution as the procuring activity specifies.

3.4.4.3 Approvals. - Unless otherwise specified by this document or the procuring activity all reports are submitted for information only.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection Methods

4.2.1 Examination. - The mass property control data shall be thoroughly examined to determine conformance with this specification with respect to all the requirements specified herein.

4.2.2 Delivery. - The data package shall be examined to ascertain that the preparation for delivery conforms to this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A. - The mass properties control data shall be preserved and packaged in accordance with Method 1C of Specification MIL-P-116.

5.1.2 Level C. - Preservation and Packaging shall be as stipulated in Paragraph 5.1.1.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T12:22Z
Check the source to verify that this is the currentversion before use.

5.2 Packing

5.2.1 Level A. - Mass properties control data shall adequately be packed to meet the carriers' rules and regulations and to insure safe delivery at destination. Such packs shall meet security regulations when contents are classified. See Paragraph 6.2.2.

6. NOTES

6.1 Intended Use. - The procuring activity's mass properties personnel will be continuously evaluating each contractor's mass properties control capability and effectiveness. The results of the evaluation will be summarized and documented within the procuring activity as frequently as conditions (overweight, incomplete reports, late submittals, etc.) warrant. However, each contractor will be completely evaluated within six months after being awarded a contract and annually thereafter. These reports will be for Government internal use only and will not be provided for contractor information. The evaluation will be based upon, but not limited to, the following:

(a) Trend relationship between contractor's mass properties summaries and specified mass properties requirements.

(b) Completeness of data submitted.

(c) Compatibility of measured data with estimated and calculated data.

(d) Compliance with data submittal schedules.

6.2 Information for Contracting Officer

6.2.1 Procurement Document. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable levels of preservation, packaging and packing required.

(c) Items of data required (see Paragraph 3.4).

(d) Additional Quality Assurance provision as applicable.

6.2.2 Classification of Data. - The applicable security regulations of the procuring activity shall control the processing and distribution of classified data.

6.3 Definitions

6.3.1 Achievable Weight. - A nominal weight value that can be reasonably expected.

6.3.2 Angles of Intersection. - The angles between the geometric axes and the principal axes of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

6.3.3 Critical Mass Properties. - Those mass properties which have limits.

6.3.4 Current Mass Properties. - The current prediction of the final nominal mass properties based on latest analytical or measured data, including appropriate allowances for growth still to be encountered, and estimates for known changes not yet formally defined.

6.3.5 Conceptual Phase. - The first phase of the System Acquisition Life Cycle. It represents the period extending from the determination of a needed operational capability to the Program Decision which authorizes the accomplishment of the Validation Phase for a proposed acquisition program. The objective of the Conceptual Phase is to define and select the system(s) which warrants continued development.

6.3.6 Deployment Phase. - The final phase of the System Acquisition Life Cycle. During this phase, the systems are provided to and used by operational units.

6.3.7 Full Scale Development Phase. - The third phase of the System Acquisition Life Cycle during which the system, including all of the items necessary for its support, is designed, fabricated and tested.

6.3.8 Growth. - The nominal predicted weight allowance for in-scope design changes due to manufacturing problems and development test results. It is not treated as an uncertainty in dispersion studies but may be subject to uncertainty analyses when included as part of a nominal critical mass property being so analyzed.

6.3.9 Ignition Weight. - The total weight of a stage, module or combinations thereof including propellants, fluids and items which vary in weight and location at the time of initiation of the ignition sequence.

6.3.10 Improvement Potentials. - Alternate configurations or design changes which, if incorporated into the basic design, would result in improved vehicle characteristics.

6.3.11 Interface. - Staging - A plane surface between two steps at which separation occurs during flight.

6.3.12 Item. - The vehicle, or portion thereof, such as a stage, module, engine, instrument unit, interstage or contract end item.

6.3.13 Limiting Conditions. - Any system operating, handling, transfer, acceptance or other conditions to which any of the following apply:

(a) Mass properties of the conditions have been established by contract as the allowable limits for satisfaction of performance incentive or other system objectives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

(b) Mass properties of the conditions are decisively constrained by limiting values outside of which the mandatory system objectives or functions will not be achieved under the current design or operating characteristics.

(c) Mass properties of the conditions exercise a decisive influence upon the acceptable system objectives or functions under the current design or operating characteristics.

6.3.14 Limit Nominal Weight. - The most adverse Nominal Weight that may be expected should the assumptions made in generating the Nominal Weight fail to materialize. When considered with the Root Sum Square (RSS) of the verification uncertainty and the manufacturing variation uncertainty, the item maximum weight should not be exceeded (see Figure 3).

6.3.15 Manufacturing Variation. - A random uncertainty resulting from the predicted or calculated effects of manufacturing processes of an item.

6.3.16 Mass Properties. - The physical characteristics of weight, center of gravity, mass distribution, moments of inertia and principal axes orientation.

6.3.17 Minimum Burnout Weight. - The weight at the end of effective action time for each stage where burning is to depletion of all main impulse propellants and fluids assumed to be available for expenditure.

6.3.18 Nominal Weight. - The final in-scope system weight which is realistically achievable and can be reasonably expected. It is the sum of the target weight and growth and represents the value about which weight is controlled. Its most adverse value is the Limit Nominal Weight (see Figure 3).

6.3.19 Outage. - The allowance for mean propellant-utilization system error.

6.3.20 Phase. - A period of time which denotes program progression.

6.3.21 Precision. - The degree of agreement of repeated independent measurements of a single quantity yielded by repeated applications of a measurement process under specified conditions.

6.3.22 Procuring Activity. - Any agency which maintains administrative control of a contract entered into by the agency and contractor.

6.3.23 Production Phase. - The fourth phase of the System Acquisition Life Cycle during which the system, including training equipment, spares, facilities, etc., is produced for operational use.

6.3.24 Random Error. - Error of a measurement method which fluctuates irregularly from observation to observation and which is caused by conditions that cannot be controlled completely.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

tests of a sufficient number of main and auxiliary rotor components to for fatigue

6.3.25 Reference Datum. - A theoretical plane, assumed to be exact for purposes of computation or reference, from which the location of vehicle features may be established.

6.3.26 Sliverage. - That portion of the solid propellant which is unburned at the end of action time.

6.3.27 Stage. - That portion of a vehicle consisting of a step plus all steps above (see Figure 2).

6.3.28 Step. - That portion of a vehicle which may be physically separated from other portions by disconnecting or staging interfaces.

6.3.29 Target Weight. - A realistic estimate of system weight based on all currently known or foreseeable elements comprising the system. It includes allowances for miscellaneous items that are needed to complete and install each subsystem (see Figure 3).

6.3.30 Total Gross Weight. - The weight of the vehicle after loading is complete and prior to ignition.

6.3.31 Uncertainty. - A general term for the estimated amount by which the observed or calculated value of a quantity may depart from the value accepted as true.

6.3.32 Validation Phase. - The second phase of the System Acquisition Life Cycle. Here the major program characteristics (technical, cost and schedule) are validated and refined through extensive study and analysis, hardware development or prototype testing, leading to the ratification decision which permits the Full-scale Development Phase. Analyses and/or test hardware and/or prototype hardware are the output of the Validation Phase.

6.3.33 Vehicle. - The entire assembly which is launched from the earth's surface.

6.3.34 Verification Uncertainties. - The root sum square of the random errors associated with the analysis technique and/or measurement technique and inventory.

6.3.35 Weight Contingency. - The nominal predicted weight allowance for deficiencies in estimated or calculated weights resulting from lack of detail in current design data. It is not treated as an uncertainty in dispersion studies but may be subjected to uncertainty analysis when included as a part of a nominal Critical Mass Property being so analyzed.

6.4 Identification of Changes to Previous Issue. - Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

Custodians:

Air Force - 19

NASA - MAP

Preparing Activity:

Air Force - 19

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

Program Phase	CONCEPTUAL AND/OR FUNDED STUDIES		VALIDATION AND/OR FULL SCALE DEVELOPMENT																			PRODUCTION AND/OR DEPLOYMENT	
			STUDIES AND ANALYSES		HARDWARE DEVELOPMENT (OR PROTOTYPE) TESTING																		
Report Type					Detail							Status					Procedural		Miscellaneous				
Column No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	
Schedule of Submittals	Semiannually for studies of 1 year or more; at completion of all studies	Every other week for studies of 2 months or less, at first of each month for studies of more than 2 months	Semiannually for studies of 1 year or more; at completion of all studies, with submittal of all proposals; at mid/or Design Reviews	Every other week for studies of 2 months or less; at first of each month for studies more than 2 months (to be continued through contract definition)	Program Plan - Definition Phase - 30 days after (or Go-Ahead) (FOR APPROVAL)	Proposed Statement of Work and Program Plan for Subsequent Phase - at completion of studies	At PDR, or within 30 days of approved definitive contract (or Go-Ahead) (FOR APPROVAL)	Annually for contracts exceeding 2 years	At CDR, or within 30 days after 90% completion of original drawing release for contracts of less than 2 years	At validation review or within 30 days after weights, or flight, specified by the procuring activity	Weight - (first of each month after approved definitive contract (or Go-Ahead) and ending with delivery of last item	Pre-flight - 120 days prior to launch	Pre-flight - 14 days after acceptance weighing	Pre-flight - 14 days after significant mass properties change	Post-flight - Quick Look - within 24 hours after launch	Post-flight - Final - 30 days after launch	Program Plan - within 30 days after definitive contract (or Go-Ahead) (FOR APPROVAL)	Verification Plan - within 6 months after definitive contract (or Go-Ahead) (FOR APPROVAL)	Contract Change Proposal - with each contract change proposal	Flight and Propellant Verification - as required by Verification Plan	(Optional) Targeting Data - as required by Verification Plan	(Optional) Targeting Data - as required by verification Plan	
NUMBER	DESCRIPTION (See Table 2 for paragraph numbers)																						
1	Title Page		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
2	Table of Contents		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
3	Review for Management		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4	Organization		X	X															X				
5	Mass Properties Summary		X	X		X			X	X	X	X		X		X			X				
6	(Intel) Mass Properties		X	X		X			X	X	X	X		X		X			X				
7	Change Analysis & Improvement Potential		X		X		X		X	X	X	X	X	X		X			X				
8	Unresolved Problems			X		X					X	X	X	X		X			X				
9	Historical Log									X		X	X	X		X			X				
10	Requested Data		X		X		X		X	X	X	X	X	X		X							
11	Powered Flight Data		X		X		X		X	X	X	X	X	X		X							
12	Diagrams		X	X	X		X		X	X	X	X	X	X		X							
13	Refurbishing Data		X		X		X		X	X	X	X	X	X		X							
14	Government Furnished Equipment		X		X		X		X	X	X	X	X	X		X							
15	Computer Cards or Tapes		X		X		X		X	X	X	X	X	X		X							
16	Mass Distribution		X		X		X		X	X	X	X	X	X		X							
17	Design Data for Mass Properties Analysis		X		X		X		X	X	X	X	X	X		X							
18	Actual Data Records		X		X		X		X	X	X	X	X	X		X							
19	Summary of Critical Mass Properties			X		X							X			X			X				
20	Evaluation of Flight				X		X		X	X	X	X	X	X		X			X				
21	Uncertainties				X		X		X	X	X	X	X	X		X			X				
22	Limits		X		X		X		X	X	X	X	X	X		X			X				
23	Fluid and Propellant Loads		X		X		X		X	X	X	X	X	X		X			X				
24	Capacity and Loading Information		X		X		X		X	X	X	X	X	X		X			X				
25	Verification Plan																						
26	Operational Targeting Data																						
27	Proposed Statement of Work																						
28	Program Plan		X		X		X		X	X	X	X	X	X		X							
29	Trade Study Data			X		X																	
30	Growth Allowance Depletion		X		X		X		X	X	X	X	X	X		X							
31	Data Sources		X		X		X		X	X	X	X	X	X		X							

NOTES:
- update only
(X) submitted as required by procuring activity

TABLE I. REPORT CONTENT AND SUBMITTAL SCHEDULE

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

(This page intentionally left blank)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

7. APPENDIXES AND INDEXES

7.1 Index

1.	SCOPE	1
1.1	General	1
1.2	Limitations	1
2.	APPLICABLE DOCUMENTS	1
2.1	Effective Issue	1
3.	REQUIREMENTS	1
3.1	Objectives	1
3.1.1	General	1
3.1.2	Development of Objectives	1
3.1.3	Development of Limits	2
3.2	Studies, Planning, Cognizance and Control	2
3.2.1	Studies	2
3.2.1.1	Trade Studies Methodology	2
3.2.2	Planning	2
3.2.2.1	Program Plan	2
3.2.2.2	Supporting Plans	2
3.2.2.3	Plan Review	2
3.2.2.4	Coordination	2
3.2.2.5	Personnel	2
3.2.2.6	Planned Statement of Work	2
3.2.2.7	Growth Depletion	3
3.2.3	Cognizance	3
3.2.3.1	Monitoring	3
3.2.3.2	Uncertainty Analyses	3
3.2.3.3	Subcontractor Surveillance	3
3.2.3.4	Associate Design Activity and GFE Suppliers Interfaces	3
3.2.3.5	Supplemental Data	3
3.2.3.6	Trade Studies Inputs	3
3.2.3.7	Identification of Changes	3
3.2.3.8	Review Meeting Support	4
3.2.3.9	Relationships	4
3.2.3.10	Interface with Associate and Integrating Contractors	4
3.2.3.11	System Engineering	4
3.2.4	Control	4
3.2.4.1	System Mass Properties Base	4
3.2.4.1.1	Specified Base	4
3.2.4.1.2	Specification Base	4
3.2.4.1.3	Reference Base	4
3.2.4.2	Improvement Action	5
3.2.4.2.1	Means to Meet Specified Mass Properties	5
3.2.4.2.2	Improvement Potentials	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.2.4.2.3	Problem Definition	5
3.2.4.2.4	Corrective Action	5
3.2.4.2.5	Limiting Condition Analyses	5
3.2.4.2.6	Document Release	5
3.3	Verification	5
3.3.1	Verification of Critical Mass Properties	5
3.3.2	Contract End Item Verification	5
3.3.3	Detail Part Verification	6
3.3.4	Measurement Conditions	6
3.3.5	Notification of Measurement	6
3.3.6	Data Records	6
3.3.7	Post Flight Analyses	6
3.3.8	Field Support of Launch Operations	6
3.4	Documentation	6
3.4.1	Types of Reports Required	6
3.4.1.1	Detail Mass Properties Reports	7
3.4.1.2	Mass Properties Status Reports	7
3.4.1.3	Procedural Reports	7
3.4.1.4	Miscellaneous Reports	7
3.4.1.4.1	Contract Change Proposals	7
3.4.1.4.2	Test Results	7
3.4.1.4.3	Operational Targeting Data	7
3.4.2	Report Format	7
3.4.2.1	General	7
3.4.2.1.1	Unit System of Measure	7
3.4.2.1.2	Forms	7
3.4.2.1.3	Coding	7
3.4.2.1.4	Coordinate Axes	8
3.4.2.2	Report Elements	8
3.4.3	Report Element Description	8
3.4.3.1	Title Page	8
3.4.3.2	Table of Contents	8
3.4.3.3	Review for Management	8
3.4.3.3.1	Management Review Tabulations	8
3.4.3.4	Organization	9
3.4.3.5	Mass Properties Summary	9
3.4.3.6	Detail Mass Properties	10
3.4.3.7	Change Analysis and Improvement Potentials	10
3.4.3.7.1	Mass Properties Change Analysis	10
3.4.3.7.2	Pending Mass Properties Change Analysis	11
3.4.3.7.3	Summary of Improvement Potentials	12
3.4.3.8	Unresolved Problems	12
3.4.3.9	Historical Log	12
3.4.3.10	Sequenced Data	12
3.4.3.11	Powered Flight Data	13
3.4.3.12	Diagrams	13
3.4.3.13	Substantiating Data	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

3.4.3.14	Government Furnished Equipment	14
3.4.3.15	Computer Cards or Tapes	14
3.4.3.16	Mass Distribution	14
3.4.3.17	Design Data for Mass Properties Analysis	14
3.4.3.17.1	Mass Properties Dependent Design Information	14
3.4.3.17.2	Structural Increments for Design Features	14
3.4.3.18	Actual Data Records	14
3.4.3.18.1	General	14
3.4.3.18.2	Records for Dry Mass Properties Measurements	15
3.4.3.18.3	Records for Fluid Mass Properties Measurements	15
3.4.3.19	Summary of Critical Mass Properties	15
3.4.3.20	Evaluation of Flight	16
3.4.3.21	Uncertainties	16
3.4.3.22	Limits	16
3.4.3.23	Fluid and Propellant Loads	17
3.4.3.24	Capacity and Loading Information	17
3.4.3.25	Verification Plan	17
3.4.3.26	Operational Targeting Data	18
3.4.3.27	Proposed Statement of Work	18
3.4.3.28	Program Plan	18
3.4.3.29	Trade Study Data	18
3.4.3.30	Growth Allowance Depletion	19
3.4.3.31	Data Sources	19
3.4.4	Report Submittal Requirements	19
3.4.4.1	Schedule of Submittals	19
3.4.4.2	Distribution	19
3.4.4.3	Approvals	19
4.	QUALITY ASSURANCE PROVISIONS	19
4.1	Responsibility for Inspection	19
4.2	Inspection Methods	19
4.2.1	Examination	19
4.2.2	Delivery	19
5.	PREPARATION FOR DELIVERY	19
5.1	Preservation and Packaging	19
5.1.1	Level A	19
5.1.2	Level C	19
5.2	Packing	20
6.	NOTES	20
6.1	Intended Use	20
6.2	Information for Contracting Officer	20
6.2.1	Procurement Document	20
6.2.2	Classification of Data	20
6.3	Definitions	20
6.3.1	Achievable Weight	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

6.3.2	Angles of Intersection	20
6.3.3	Critical Mass Properties	21
6.3.4	Current Mass Properties	21
6.3.5	Conceptual Phase	21
6.3.6	Deployment Phase	21
6.3.7	Full Scale Development Phase	21
6.3.8	Growth	21
6.3.9	Ignition Weight	21
6.3.10	Improvement Potentials	21
6.3.11	Interface	21
6.3.12	Item	21
6.3.13	Limiting Conditions	21
6.3.14	Limit Nominal Weight	22
6.3.15	Manufacturing Variation	22
6.3.16	Mass Properties	22
6.3.17	Minimum Burnout Weight	22
6.3.18	Nominal Weight	22
6.3.19	Outage	22
6.3.20	Phase	22
6.3.21	Precision	22
6.3.22	Procuring Activity	22
6.3.23	Production Phase	22
6.3.24	Random Error	22
6.3.25	Reference Datum	23
6.3.26	Sliverage	23
6.3.27	Stage	23
6.3.28	Step	23
6.3.29	Target Weight	23
6.3.30	Total Gross Weight	23
6.3.31	Uncertainty	23
6.3.32	Validation Phase	23
6.3.33	Vehicle	23
6.3.34	Verification Uncertainties	23
6.3.35	Weight Contingency	23
6.4	Identification of Changes to Previous Issue	23

Figure 1	Vehicle Coordinate Axes and Notation System	32
Figure 2	Sample Missile Step-Stage Relationship	33
Figure 3	Weight Relationships	33

Table 1	Report Content and Submittal Schedule	25
Table 2	Report Element - Tasks Relationship	34

7. APPENDIXES AND INDEXES

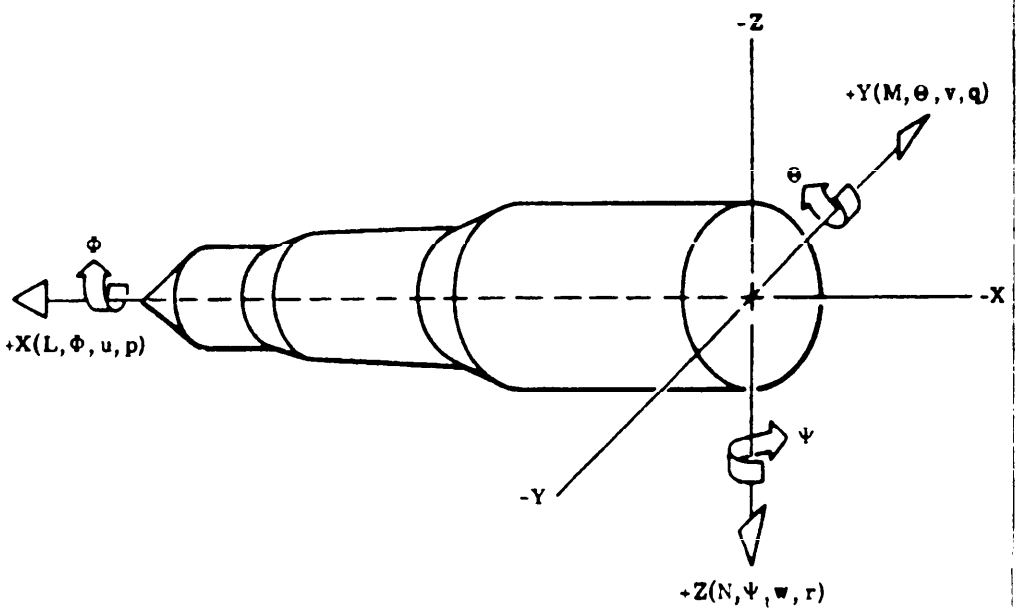
7.1	Index	27

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

7.2 Appendixes

Appendix A - Reporting Forms A-1
Appendix B - Functional Code. B-1
Appendix C - Design Data List C-1
Appendix D - Parameters and Inventory of Fluids and Propellants Loaded . D-1
Appendix E - Mass Properties Verification Plan E-1
Appendix F - Capacity and Loading Information for Fluids and Propellants. F-1

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.



DIRECTION	AXIS SYMBOL	MOMENT SYMBOL	POSITIVE ROTATION	ANGLE	ANGLE SYMBOL	LINEAR VELOCITY	ANGULAR VELOCITY
LONGITUDINAL	X	L	Y TO Z	ROLL	Φ	u	p
LATERAL	Y	M	Z TO X	PITCH	Θ	v	q
VERTICAL	Z	N	X TO Y	YAW	Ψ	w	r

THE RIGHT-HAND RULE IS THE BASIS FOR THE SPACE VEHICLE SIGN CONVENTION

Figure 1. Vehicle Coordinate Axes and Notation System

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

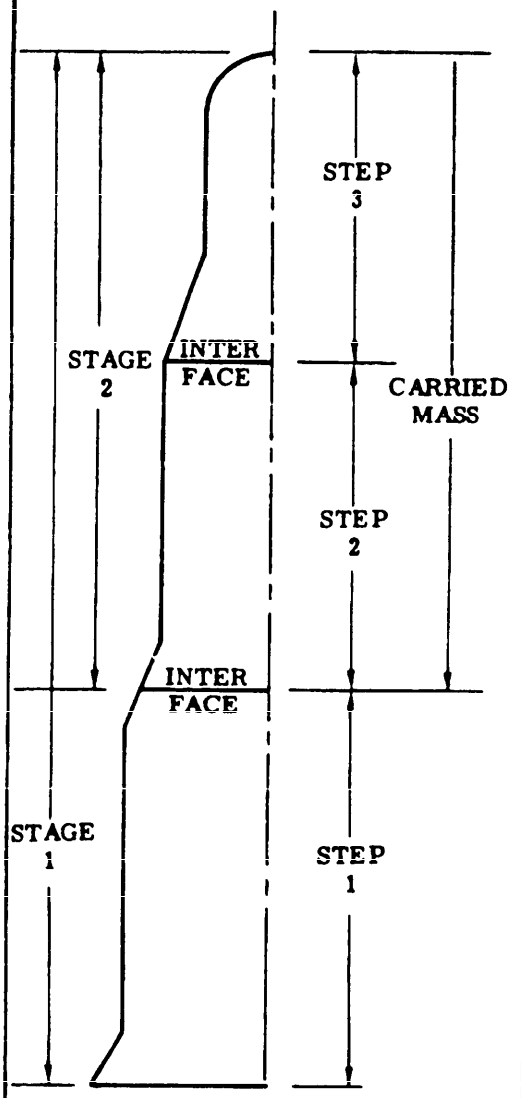


FIGURE 2

SAMPLE

MISSILE STEP-STAGE RELATIONSHIP

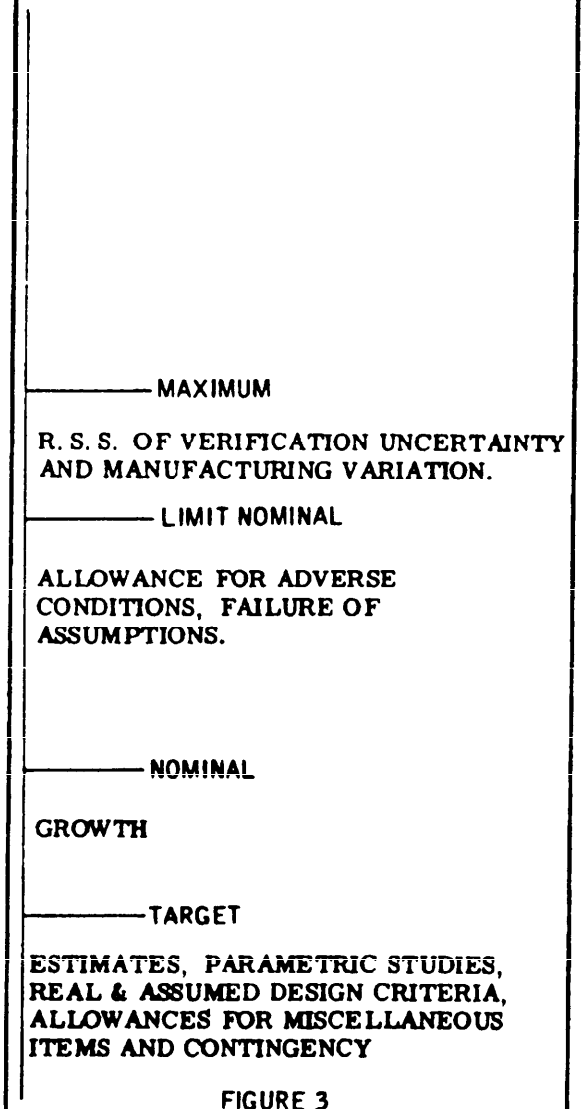


FIGURE 3

WEIGHT RELATIONSHIPS

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

REPORT ELEMENT DESCRIPTION	PARAGRAPH NUMBER	ASSOCIATED TASK(S) PARAGRAPH NUMBER(S)
Title Page	3.4.3. 1	
Table of Contents	2	
Review for Management	3	
Organization	4	3.2.2.5
Mass Properties Summary	5	3.2.3; 3.2.4
Detail Mass Properties	6	3.2.3; 3.2.4
Change Analysis & Improvement Potentials	7	3.2.3.7; 3.2.4.2
Unresolved Problems	8	3.2.4.2.3; 3.2.4.2.4
Historical Log	9	3.2.3; 3.3.6
Sequenced Data	10	3.2.3
Powered Flight Data	11	3.2.3
Diagrams	12	3.2.3.5
Substantiating Data	13	3.1.2
Government Furnished Equipment	14	3.2.3.4
Computer Cards or Tapes	15	3.2.3.5
Mass Distribution	16	3.2.3
Design Data for Mass Properties Analysis	17	3.1.2; 3.2.3
Actual Data Records	18	3.3
Summary of Critical Mass Properties	19	3.1.2; 3.2.3; 3.4.2.5
Evaluation of Flight	20	3.2.3; 3.3.7
Uncertainties	21	3.1.2; 3.2.3.2
Limits	22	3.1.2; 3.1.3; 3.2.3; 3.4.2.5
Fluid and Propellant Loads	23	3.2.3
Capacity and Loading Information	24	3.2.3
Verification Plan	25	3.2.2; 3.3
Operational Targeting Data	26	3.2.2.2; 3.2.3; 3.3
Proposed Statement of Work	27	3.2.2.6
Program Plan	28	3.2.2
Trade Study Data	29	3.2.1; 3.2.3.6
Growth Allowance Depletion	30	3.2.2.7; 3.2.3
Data Sources	3.4.3.31	

TABLE 2. REPORT ELEMENT/TASKS RELATIONSHIP

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

This appendix contains sample forms to be used as a guide in the construction and use of forms as discussed in paragraph 3.4.2.1.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

MASS PROPERTIES SUMMARY										
CODE	DESCRIPTION	SPECIFIED WEIGHT BASE ( )	PROCURING ACTIVITY AND GFE CHANGES	REVISED SPECIFIED WEIGHT BASE	CURRENT WEIGHT	CHANGES LAST TO CURRENT	PERCENTAGE BREAKDOWN OF CURRENT WEIGHT			NOTE NUMBER
							EST	CALC	ACT	

PREPARED BY _____ IDENTIFICATION REPORT NO. _____
DATE _____ PAGE ___ OF _____

FORM 1 PART 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

MASS PROPERTIES SUMMARY

CODE	DESCRIPTION	CURRENT WEIGHT ( )	CENTER OF GRAVITY (____ FROM REF. DATUM)			MOMENT OF INERTIA (____)		
			X	Y	Z	PITCH	ROLL	YAW

PREPARED BY _____ IDENTIFICATION
DATE _____

REPORT NO. _____
PAGE ____ OF ____

FORM 1 PART II

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

DETAIL MASS PROPERTIES

CODE	DESCRIPTION	CLASS (E, C, A)	WEIGHT (_____) CODE GENERATION FIRST	SECOND	THIRD	CENTER OF GRAVITY (__ FROM REF. DATUM) X	Y	Z

IDENTIFICATION

PREPARED BY _____
DATE _____

REPORT NO. _____
PAGE ___ OF ___

FORM 2 PART I

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

DETAIL MASS PROPERTIES									
CODE	DESCRIPTION	CURRENT WEIGHT ( ) CODE GENERATION			MOMENT OF INERTIA ( )			NOTE NUMBER
		FIRST	SECOND	THIRD	PITCH	ROLL	YAW	

PREPARED BY _____ IDENTIFICATION REPORT NO. _____
DATE _____ PAGE ___ OF _____

FORM 2 PART II

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

MASS PROPERTIES CHANGE ANALYSIS										
*CHANGE IDENTIFICATION AND/OR *NOTE NUMBER	CODE	EFFECTIVE POINT	WEIGHT CHANGE ( )			CENTER OF GRAVITY (__ FROM REF. DATUM)			MOMENT OF INERTIA ( )		
			TOTAL	RESPONSIBILITY		X	Y	Z	PITCH	ROLL	YAW
				DESIGN ACTIVITY	PROCURING ACTIVITY						

PREPARED BY _____ IDENTIFICATION REPORT NO. _____
DATE _____ PAGE ____ OF _____

FORM 3 PART I *NOTE: Cross out if not applicable

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

MASS PROPERTIES CHANGE ANALYSIS			
*CHANGE IDENTIFICATION AND/OR *NOTE NUMBER	CODE	TOTAL WEIGHT CHANGE (_____)	REMARKS

PREPARED BY _____ IDENTIFICATION REPORT NO. _____
DATE _____ PAGE ____ OF _____

FORM 3 PART II *NOTE: Cross out if not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

WEIGHT AND BALANCE HISTORY LOG

Sheet No. _____

Place an X in blanks not required for identification

ITEM

SPACE VEHICLE _____ S/C MODULE _____ ENGINE MODEL _____

MISSION _____ L/V STAGE _____ SERIAL NO. _____

ENTRY NO.	DATE / TIME	SIGNATURE	ITEM DESCRIPTION (DWG NO, SERIAL NO, NAME, SYSTEM)	CODE	+ / -	WEIGHT (____)	CENTER OF GRAVITY (____ FROM REF DATUM)		
							X	Y	Z

FORM 4 PART I

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

WEIGHT AND BALANCE HISTORY LOG

Sheet No. ____

WORK SHEET

Place an X in blanks not required for identification

ITEM

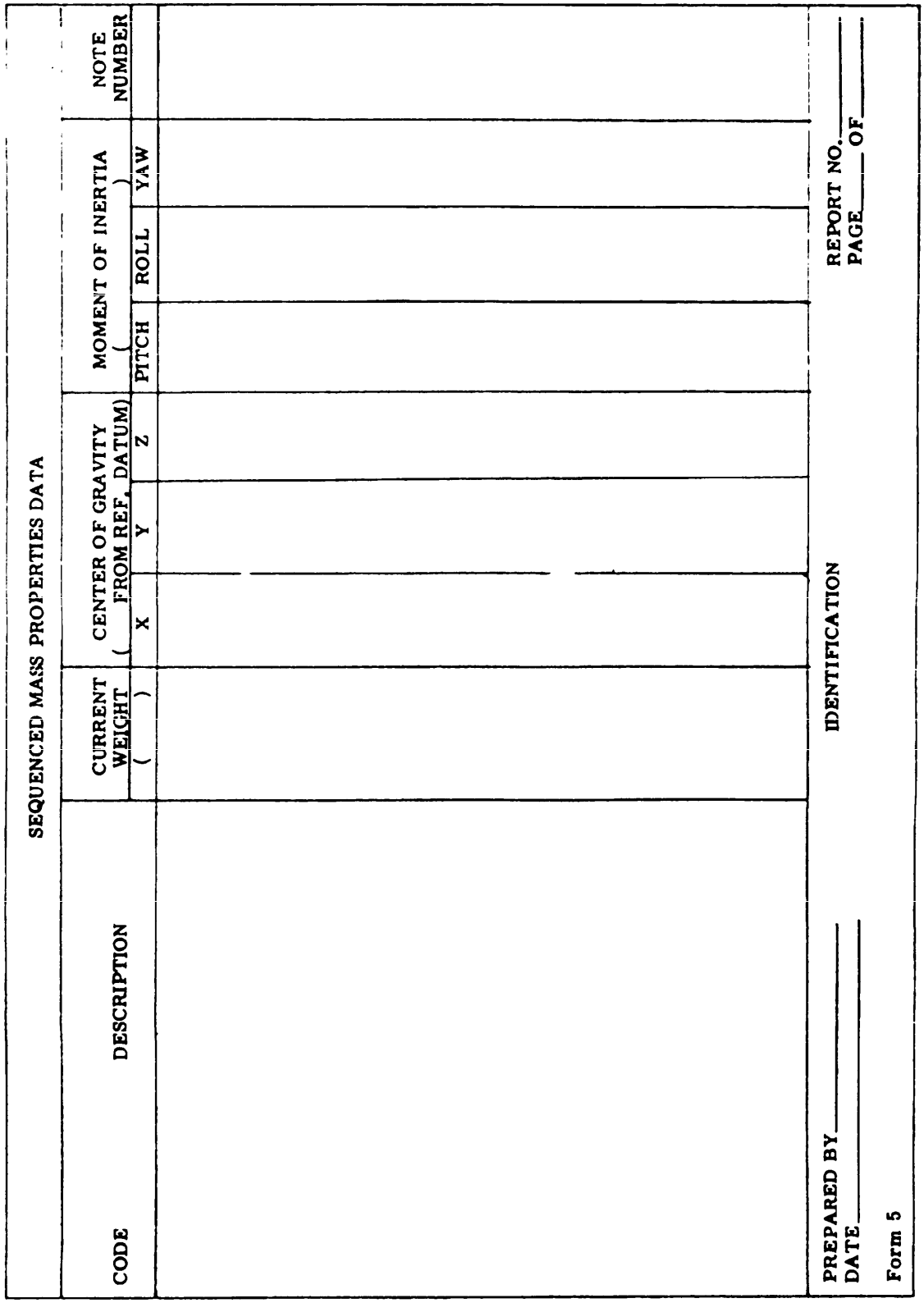
SPACE VEHICLE _____ S/C MODULE _____ ENGINE MODEL _____

MISSION _____ L/V STAGE _____ SERIAL NO. _____

ENTRY NO.	+ / -	WEIGHT (____)	CENTER OF GRAVITY			PRODUCT OF INERTIA			TRANSFER FACTORS			MOMENT OF INERTIA ( )			MOMENT OF INERTIA ( )		
			X	Y	Z	WXY	WXZ	WYZ	X^2	Y^2	Z^2	Ioy	Iox	Ioz	PITCH	ROLL	YAW

FORM 4 PART II

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

APPENDIX B

FUNCTIONAL CODE

B. 10. 0 OBJECTIVES

The functional code provides a coding system for a three-generation breakdown of vehicle items according to their functional use. The objectives of the functional code are:

a. Provide a basis for computing weight summaries.

b. Allow direct substantiation of weight summaries and analysis methods.

c. Provide a uniform basis for design weight comparison of vehicle systems.

d. Facilitate the preparation of weight summaries for complete vehicles such that a given section, stage, or module summary may be readily included in the summary of the total vehicle.

e. Provide identifiable vehicle coordinate location data.

B. 20. 0 GENERAL DESCRIPTION

The functional code, along with the nomenclature employed, shall be considered as the basic functional breakdown of vehicle items. Using this code, any type of vehicle may be coded for the purpose of mass properties reporting. The code consists of first generation items, each of which is broken down into second generation items. The second generation items are further broken down into third generation items. The first 16 first-generation codes include items which are essentially fixed in location and weight, the summary of which is dry weight. The remaining first generation codes include items which are variable either in location or weight. The functional code is intended to be complete, with the exception of the third generation breakdown. The third generation items listed in the code are intended to serve as a guide and do not necessarily represent a complete or fixed third generation breakdown. Further structural and equipment breakdowns, and the generous use of footnotes, are encouraged when such practices will further the understanding of tabulated data and aid in the intended purposes. The use of the words "provision" and "installation" shall be avoided unless further breakdown is given to indicate what is included in the item.

B. 30. 0 USAGE

All coding shall use the functional code system. Each identifiable section, stage or module shall be coded separately. When coding items, the design activity shall not deviate from the first generation breakdown. When necessary, second generation items may be added, if prior approval has been obtained from the procuring activity. The third generation breakdown identifies the second generation, and may be modified or expanded as required as long as the identity of the second generation is not changed and sufficient detail is provided such that a multistation analysis may be validated. It may be practical in some instances to code complete assemblies, but in such cases, the items making up the assembly shall not be coded. Once an item has been coded, the code shall not be changed without the approval of the procuring activity.

B. 30. 1 Items Jettisoned During Flight. - All items allocated to dry weight that are jettisoned or ablated during flight shall be noted and identified.

B. 30. 2 Attachment and Support Items. - All loose items of attachment equipment and system supports (nuts, bolts, rivets, etc.), shall be allocated to the same group as the item being attached or supported. Multipurpose supports shall be allocated to the primary function. Where a primary allocation cannot be determined by observation, such as may occur for fairings, fairleads, supports for wire harness, brackets, etc., an arbitrary allocation may be made.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

B. 30. 3 Dual Allocations. - Where provisions are made in both dry weight and variable load categories for a given installation, allocations shall be made to the dry weight or variable load in accordance with the category allocation assigned in the model or detail specification or in the general design specification and/or instructions of the procuring activity. If a conflict in category allocation of any items occurs between the weight statement and model or detail specification, the latter shall govern in all cases.

B. 30. 4 Power Distribution Systems. - All items of power distribution systems (electrical, hydraulic, and pneumatic), including fluid in the respective tubing for hydraulic systems, from main distribution points to actuating units shall be allocated with the items operated. All components from source of power to main distribution points shall be considered to comprise the main system. The distribution system for a given item shall be assumed to include all components which would be unnecessary and could be removed if power actuation of the item were to be eliminated. Where several successive branches from a distribution system are involved, and a clear-cut main distribution point is not readily apparent, it shall be assumed that power is supplied to a given geographic location primarily for the function requiring the largest power, and to other functions in the sequence of descending power requirements.

B. 30. 5 Items Performing Multipurpose Functions. - Items which cannot be isolated as serving a single function, and which are integral with the structure of aerodynamic surfaces or body, shall be allocated to the appropriate structural group. Typical in this category are:

a. Ducting for air supply, bleed air and exhaust in which skin or webs serve as part of the duct contour. The duct wall and stiffeners which can be isolated primarily as duct systems are allocated to Main Propulsion - Air Breathing - - Air Induction System or to Environmental Control.

b. Integral tanks in which skin, webs, solid bulkheads and load carrying seal fittings are utilized to form the cavity. The separate tank supports, sealant material, non-load carrying fittings are allocated to "Main Propulsion."

c. Skin, ablative material or special material integral with the structure and required for adverse environments, including elevated temperature, shall be included in the structural groups. Nonstructural temperature control systems for structure are allocated to "Induced Environment Protection."

d. Radomes which do not form normal contour of the aerodynamic surface or body are allocated to the "Guidance and Navigation" or "Communication" groups.

e. Removable structural panels or structural doors for special functions are allocated to the basic structural groups.

B. 30. 6 Items Common to Several Structural Groups. - Items common to several structural groups consist of:

a. Joints, Splices and Fasteners. These items include all doublers, fillers, spacers, gussets, angles, clips, rivets, bolts, nuts, and screws that should, insofar as practicable, be allocated to the structural function served.

b. Bulkheads, Frames, Cover, and Longerons. Bulkheads are the main structural members which assist in carrying loads around major cutouts, or which distribute major internal or external concentrated loads and may also act as partitions to separate an area into physical compartments. Body bulkheads should be listed individually and identified by location. Frames are similar to bulkheads but usually of relatively lighter construction and serve the function primarily of stiffening the cover or skin to maintain contour under load. Longerons distribute the load longitudinally, along with longitudinal partitions which contribute to body strength.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

c. Exterior Finish and Sealer. This includes paint, finish, and sealer for protection from the elements, and is coded to the item it covers.

B. 40. 0 FIRST GENERATION LISTING AND DESCRIPTION

The following list presents all of the first generation functional code items and a brief description of each. These descriptions shall be used to establish the coding of space vehicle systems and components.

B. 40. 1 Aerodynamic Surfaces. - The basic and secondary load-carrying members for all primary lifting and aerodynamic control surfaces, both fixed and movable, exclusive of the nonstructural panels used for induced environmental protection systems. The various surfaces are identified as:

a. Leading edge: all structure forward of the front spar.

b. Trailing edge: all structure aft of the rear spar.

Surface-body splices and fittings are to be allocated to the surface.

B. 40. 2 Body Structure. - The basic and secondary load-carrying members, exclusive of the nonstructural panels used for induced environmental protection systems. Identify, by note, the structure that is jettisoned during flight.

B. 40. 3 Induced Environment Protection. - The devices which in themselves, or in combination, protect the missile or space vehicle structure from the detrimental effects of heat, noise, meteorites and radiation.

B. 40. 4 Launch, Recovery and Docking. - Items that provide the vehicle with the capability to be launched from or brought to rest with respect to a mass. Enter descriptive or location data, as appropriate, for clarification of the function served.

B. 40. 5 Main Propulsion. - Propulsive items which provide flight path thrust and acceleration and include rocket engines, nuclear engines, propulsive devices, and related equipment, such as fuel systems, oxidizer systems, and pressurizing systems.

Other propulsion devices, such as those used for electrical power or to maintain control of orientation, are coded elsewhere. Exception -- thrust vectoring systems which are integral with the engine are allocated to the engine.

a. Fluid and Gas Systems Control - list liquid level sensors and any control system, such as propellant-loading gaging system and propellant utilization system, which senses propellant tank levels and transmits appropriate control signals to the engine.

b. Basic Power Plant Control - list controls, including valves which are associated with the start, operation and shutdown of the power plant. These controls are usually supplied by the power plant manufacturer.

c. Solid Propellant Engine and Accessories, Inner and Outer Insulation - list the total weight of each, including that which is burned.

d. Nuclear and Air Breathing Power Plants - (1) Nuclear shielding: list all shielding required for reactor functioning. Shielding for environmental protection is coded in Induced Environmental Protection group. (2) Air breathing: list all air breathing systems such as nacelle, pylon, and engine section, with an adequate structural breakdown (skin, bulkhead, etc.), provided a clear distinction can be made (i.e., not integral with the body structure).

Source: https://assist.dla.mil -- Downloaded: 20261 0-21T12:22Z
Check the source to verify that this is the current version before use.

e. Propellant System - list the items that contain and feed the needed propellants to the propulsion unit. As a minor deviation in the normal functional subdivision of the vehicle, the lubrication systems are included with the propellant system.

B. 40. 6 Orientation Controls, Separation and Ullage. - Control units, exclusive of navigation and guidance, which provide relatively small amounts of thrust or force compared to the main propulsion system, usually for purposes such as velocity control, attitude control, ullage control, rendezvous, and docking. Aerodynamic and spatial controls include the electro-mechanical, hydraulic, or pneumatic actuation system, from the actuator source to the item actuated.

B. 40. 7 Prime Power Source. - Systems used to generate power for purposes other than propulsion, including the source of initial power. However, once the power has been generated, any additional conversion equipment is listed under the Power Conversion and Distribution group. Indicate if the power source includes an integral pump or generator. All separated pumps or generators are to be included in the Power Conversion and Distribution group (i. e., hydraulic, pneumatic, or electrical).

B. 40. 8 Power Conversion and Distribution. - Systems used to distribute electrical, hydraulic, or pneumatic power. The source of initial power is included in the Prime Power Source group.

B. 40. 9 Guidance and Navigation. - This group is divided into the following major subgroups:

a. Guidance Source - receives a sensor signal.

b. Guidance Evaluation - evaluates signals, determines navigation requirements and informs the output systems.

c. Output - activates the control systems.

B. 40. 10 Instrumentation. - Measuring, signal conditioning, recording, and programming systems for data sampling and recording, including the sensors, circuitry, signal converters, and recording media from the measurement source to a point of telemetry or permanent storage.

B. 40. 11 Communication. - The equipment required for all means of communication within, emanating from, and received by the missile or space vehicle. This includes such items as transmitters, receivers, antennas, power amplifiers, television cameras, and spares.

B. 40. 12 Environmental Control. - Controls internal environmental conditions such as temperature, pressure, humidity, atmospheric constituents, and odor for personnel and equipment.

B. 40. 13 Armament. - Provides for the allocation of items functioning or supporting the primary armament listed in the missile load.

B. 40. 14 Personnel Provisions. - Items within the crew cabin, such as accommodations, fixed life support equipment, cargo handling, furnishings and built-in emergency equipment.

B. 40. 15 Crew Station Controls and Panels. - Items consisting of crew station controls, pedestals, stands, and display panels for all systems. Actuation of the controls may be accomplished manually or in combination with electronic, hydraulic or pneumatic devices.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

B. 40. 16 Range Safety and Abort. - Systems that indicate malfunctions which will endanger personnel or damage equipment. These systems may also initiate remedial action automatically or perform upon command for emergency conditions detected by the system.

B. 40. 17 Personnel. - The crew required to perform a particular mission, including the nonfixed items required to support the crew both inside and outside of the spacecraft, such as personal gear, life support items, and crew accessories.

B. 40. 18 Cargo. - Items stored aboard the spacecraft that will be required to perform certain functions during the mission. These items include scientific instruments and equipment to perform experiments, passengers and associated equipment.

B. 40. 19 Ordnance. - The materials and their immediate containers whose normal function is to detonate or explode.

B. 40. 20 Ballast. - Temporary ballast within the vehicle for purposes of trim. This ballast may be in the form of liquids or solids. Fixed ballast shall be included in dry weight, functional code 2. 13.

B. 40. 21 Residual Propellant and Service Items. - Propellant and service items, remaining in an item, which are not usable.

B. 40. 22 Reserve Propellant and Service Items. - Propellant and service items carried by a propulsion stage or module in excess of that required to perform a mission with a nominal vehicle.

B. 40. 23 Inflight Losses. - Propellant losses associated with the use of auxiliary propulsion systems. These include losses for roll and attitude control, venting losses from pressurization gases, and boil-off losses.

B. 40. 24 Thrust Decay Propellant. - Propellant consumed from the specified value of thrust, following the engine cutoff signal, to stage or module separation or to zero value of thrust.

B. 40. 25 Full Thrust Propellant. - Propellant consumed during the burning period from the specified value of thrust following ignition, or from lift-off from the pad, to the specified value of thrust following the cutoff signal.

B. 40. 26 Thrust Buildup Propellant. - Propellant consumed from ignition to the specified value of thrust, or consumed prior to lift-off from the launch pad.

B. 40. 27 Pre-ignition Losses. - Fluid losses associated with starting the primary propulsion system of a stage or module which occur prior to the ignition signal.

B. 50. 0 FUNCTIONAL CODE LISTING

The following pages list the first, second, and third generation breakdown of vehicle items. As discussed in par. B. 20. 0, the third generation items are intended to serve as a guide and do not necessarily represent a complete or fixed third generation breakdown.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST-GENERATION LISTING

1. Aerodynamic Surfaces
2. Body Structure
3. Induced Environment Protection
4. Launch, Recovery and Docking
5. Main Propulsion
6. Orientation Controls, Separation and Ullage
7. Prime Power Source
8. Power Conversion and Distribution
9. Guidance and Navigation
10. Instrumentation
11. Communication
12. Environmental Control
13. Armament
14. Personnel Provisions
15. Crew Station Controls and Panels
16. Range Safety and Abort
17. Personnel
18. Cargo
19. Ordnance
20. Ballast
21. Residual Propellant and Service Items
22. Reserve Propellant and Service Items
23. Inflight Losses (Liftoff to Separation)
24. Thrust Decay Propellant
25. Full Thrust Propellant
26. Thrust Propellant Buildup
27. Pre-Ignition Losses

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	1. AERODYNAMIC SURFACES
SECOND GENERATION	THIRD GENERATION
1.1 Fixed Surfaces	Basic structure Leading edge Trailing edge Secondary structure Trim devices Insulation Exterior finish and sealer
1.2 Movable Surfaces	Basic structure Leading edge Trailing edge Secondary structure Trim devices Insulation Exterior finish and sealer
1.3 Fairings and Associated Structure (Including Insulation)	Forward fairings Center fairings Tail fairings Air intakes Cable conduits Ice shields Equipment fairings Exterior finish and sealer
1.19 Contingency	
1.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	2. BODY STRUCTURE
SECOND GENERATION	THIRD GENERATION
2.1 Structural Fuel Tank	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Forward bulkhead Aft bulkhead Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Container wall insulation - inner Container wall insulation - outer Forward bulkhead insulation - inner Forward bulkhead insulation - outer Aft bulkhead insulation - inner Aft bulkhead insulation - outer Antislosh devices
2.2 Structural Oxidizer Tank	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Forward bulkhead Aft bulkhead Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Container wall insulation - inner Container wall insulation - outer Forward bulkhead insulation - inner Forward bulkhead insulation - outer Aft bulkhead insulation - inner Aft bulkhead insulation - outer Antislosh devices
2.3 Structural Propellant Tank (Common Bulkhead)	Skin (Including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Forward bulkhead Common bulkhead (including insulation) Aft bulkhead Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Container wall insulation - inner Container wall insulation - outer Forward bulkhead insulation - inner Forward bulkhead insulation - outer Aft bulkhead insulation - inner Aft bulkhead insulation - outer Antislosh devices

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	2. BODY STRUCTURE (Cont'd)
SECOND GENERATION	THIRD GENERATION
2.4 Structural Solid Propellant Case or Tank	Forward closure assembly Cylinder assembly Aft closure assembly Nozzle adapter Attaching assemblies Insulation
2.5 Structure Enclosing Nonintegral Tanks	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Bulkheads Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Insulation - inner Insulation - outer Platforms and catwalks
2.6 Structure Forward of Integral Tanks	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Bulkheads Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Insulation - inner Insulation - outer Platforms and catwalks
2.7 Structure Between Integral Tanks	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Bulkheads Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Insulation - inner Insulation - outer Platforms and catwalks
2.8 Structure Aft of Integral Tanks	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Bulkheads Fittings Brackets, doublers, gussets, etc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	2. BODY STRUCTURE (Cont'd)
SECOND GENERATION	THIRD GENERATION
2.8 Structure Aft of Integral Tanks (cont'd)	Cutouts and associated structure Insulation - inner Insulation - outer Platforms and catwalks
2.9 Thrust Structure	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Bulkheads Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Hold-down posts Insulation
2.10 Interstage/Spacer/ Vehicle Instrument Unit Structure	Skin (including welds) Ring frames, stringers, longerons, struts, and tie rods Beams and shear panels Bulkheads Fittings Brackets, doublers, gussets, etc. Cutouts and associated structure Insulation - inner Insulation - outer Platforms and catwalks
2.11 Pressurized Compartment	Bulkheads Rings Webs Frames Longerons Stringers Covering (skin, including welds) Reinforcements Partitions Floor (acting as structural member) Hatches and airlocks Windows and ports Firewalls Pressure seals Brackets, doublers, gussets, etc.
2.12 Nonpressurized Compartment	Bulkheads Rings Webs Frames Longerons Stringers Covering (skin, including welds)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	2. BODY STRUCTURE (Cont'd)
SECOND GENERATION	THIRD GENERATION
2.12 Nonpressurized Compartment (cont'd)	Reinforcements Partitions Floors (acting as structural member) Hatches and airlocks Windows and ports Firewalls Pressure seals Brackets, doublers, gussets, etc.
2.13 Ballast	Ballast for longitudinal center of gravity control Ballast for radial center of gravity control Ballast for weight simulation
2.14 Multipurpose Equipment-Containers, Panels Supports	
2.15 Exterior Finish and Sealer	
2.19 Contingency	
2.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	3. INDUCED ENVIRONMENT PROTECTION
SECOND GENERATION	THIRD GENERATION
3.1 Thermal Protection (Active)	Ablator Insulation Reinforcements Release mechanism Fairings Attaching structure Bonding material
3.2 Thermal Protection (Passive)	Insulation Reinforcements Release mechanism Fairings Attaching structure Bonding material
3.3 Noise Protection	Insulation Attaching structure Bonding material
3.4 Meteorite Protection	Bumper Supports Stiffeners Filler Shield Attaching structure
3.5 Radiation Protection	Shielding Attaching structure
3.19 Contingency	
3.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	4. LAUNCH, RECOVERY, AND DOCKING
SECOND GENERATION	THIRD GENERATION
4.1 Launch Gear	Struts Pads Fittings Deploying devices Sequencing devices Controls Braces Power source Shock attenuation devices Structure Attachment fittings Tie-down fittings
4.2 Deployable Aerodynamic Devices	Drogue parachute Main parachute Pilot parachute Paraglider Structure Sequencing controls
4.3 Landing Gear	Struts Pads Fittings Deploying devices Sequencing devices Controls Braces Power source Shock attenuation devices Solid brake rocket cases Liquid brake rockets Structure
4.4 Flotation Gear	Container Flotation devices Power source Control devices Sequencing devices Deployment devices Structure
4.5 Docking Structure	Rings Seals Latching mechanisms Repositioning devices Separation devices Fairings

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	4. LAUNCH, RECOVERY, AND DOCKING (Cont'd)
SECOND GENERATION	THIRD GENERATION
4.6 Recovery Aids	Visual Communication Tracking Landing Structure
4.19 Contingency	
4.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	5. MAIN PROPULSION
SECOND GENERATION	THIRD GENERATION
5.1 Liquid Rocket Engine and Accessories	Thrust chamber Turbopump Engine mounts (attached to engine) Gimbal mounts and thrust frame (attached to engine) Propellant plumbing (attached to engine) Gas generator and igniter (less ordnance) Lubrication system Controls Hydraulic system Pneumatic system Electrical system Heat exchangers and exhaust ducts Purge system
5.2 Solid Propellant Engine and Accessories	Nozzle - fixed Nozzle - movable Propellant container (nonstructural) Engine attachment fittings Igniter (less ordnance) Controls Insulation - inner Insulation - outer Safe and arm system (less ordnance) Thrust termination system (less ordnance)
5.3 Nuclear Power Plant and Accessories	Propellant feed system (including turbopump attached to engine) Reactor (including shielding) Thrust chamber (including nozzle) Thrust structure (attached to engine) Cooling system Controls Pneumatic system Hydraulic system Destruct system (less ordnance)
5.4 Ion Engine and Accessories	Propellant feed system (attached to engine, including turbopump) Reactor (including shielding) Thrust chamber (including nozzle) Thrust structure (attached to engine) Cooling system Pneumatic system Hydraulic system Turbine Power generating system Destruct system (less ordnance)

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:22Z
Check the source to verify that this is the currentversion before use.

FUNCTIONAL CODE

FIRST GENERATION	5. MAIN PROPULSION (Cont'd)
SECOND GENERATION	THIRD GENERATION
5.5 Photon Engine and Accessories	Power source Light source Reflector Destruct system (less ordnance) Controls
5.6 Air Breathing Engine and Accessories	Propellant feed system Thrust structure Thrust chamber Turbine Compressor Controls Air induction
5.7 Purge System for Stage Chilldown	Bottles and supports Valves and nozzles Controls Plumbing and fittings Manifolds Ducts for chilldown or purge gas (including brackets and attaching hardware) Buckets or collectors for disposal
5.8 Fuel Container (Nonstructural)	Structure Baffling and antivortex webs Liners Bladders Fill and drain provisions Vents Sumps Sensors Supports Insulation
5.9 Fuel System	Pump installation (including power supply attached to engine) Fill and drain system Distribution or suction system Replenishing system (if separate) Transfer system Vent system Purge system Antivortex devices Exclusion risers Sensing controls
5.10 Pressurization System - Fuel	Tank structure Supports Fill and drain provisions Vents Sumps

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	5. MAIN PROPULSION (Cont'd)
SECOND GENERATION	THIRD GENERATION
5.10 Pressurization System - Fuel (cont'd)	Sensors Plumbing Valves and regulators Insulation Heat exchanger Sensing controls
5.11 Oxidizer Container (Nonstructural)	Structure Baffling and antivortex webs Liners Bladders Fill and drain provisions Vents Sumps Sensors Supports Insulation
5.12 Oxidizer System	Pump installation including power supply (not attached to engine) Fill and drain system Distribution or suction system Replenishing system (if separate) Transfer system Vent system Purge system (if separate) Antivortex devices Exclusion risers Sensing controls
5.13 Pressurization System - Oxidizer	Tank structures Supports Fill and drain provisions Vents Sumps Sensors Plumbing Valves and regulators Insulation Heat exchanger Sensing controls
5.14 Auxiliary Fluids System	Tanks and supports (not integral with structure) Pump installation including power supply (not attached to engine) Fill and drain system Distribution or suction system Replenishing system (if separate) Transfer system Vent system

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	5. MAIN PROPULSION (Cont'd)
SECOND GENERATION	THIRD GENERATION
5.14 Auxiliary Fluids System (cont'd)	Tank pressurization system Purge system (if separate) Antivortex devices Exclusion risers Sensing controls
5.15 Propellant Utilization System	Plumbing Valves Regulators Supports
5.16 Lubricating System	Tanks, tubing, valves, accumulators, bottles, supports, etc. Controls Power supply (if separate) Wiring, cables (including clamps, hardware, etc.)
5.19 Contingency	
5.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	6. ORIENTATION CONTROLS, SEPARATION, AND ULLAGE
SECOND GENERATION	THIRD GENERATION
6.1 Thrust System (Main)	Thruster Injection system Gimbal system (if not integral) Insulation Protective devices Case Liner Ignition system (less ordnance) Nozzles Structure
6.2 Thrust System (Auxiliary)	Thruster Injection system Gimbal system (if not integral) Insulation Protective devices Case Liner Ignition system (less ordnance) Nozzles Structure
6.3 Aerodynamic Control	Roll Pitch Yaw
6.4 Spatial Control	Roll Pitch Yaw Stabilizing electronics
6.5 Separation	Structure Rocket cases Fairings Controls, sequencing equipment, and wiring Jettison system Power supply
6.6 Ullage (Separate from 6.4, 6.5)	Structure Rocket cases Fairings Controls, sequencing equipment, and wiring
6.7 Fuel Containers	Tanks Bladders Baffles Fill and drain provisions Sensors and valves Insulation Structure

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	6. ORIENTATION CONTROLS, SEPARATION, AND ULLAGE (Cont'd)
SECOND GENERATION	THIRD GENERATION
6.8 Oxidizer Containers	Tanks Bladders Baffles Fill and drain provisions Sensors and valves Insulation Structure
6.9 Pressurization	Tanks Liners Fill and drain provisions Insulation Plumbing Sensors and valves Sequencing Structure
6.10 Distribution and Control - Fuel	Plumbing Valves and regulators (indicate injection valves) Insulation Structure
6.11 Distribution and Control - Oxidizer	Plumbing Valves and regulators (indicate injection valves) Insulation Structure
6.12 Thrust Structure	Ring Frame Supports Fittings Skin Insulation Stringers
6.13 Artificial Gravity	
6.19 Contingency	
6.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	7. PRIME POWER SOURCE
SECOND GENERATION	THIRD GENERATION
7.1 Power Source -	Engine Engine pump unit Engine generator unit Gas generator unit Gas generator Fuel storage Plumbing Valves and pumps
7.2 Power Source - Fuel Cell	Fuel cell - nonregenerative modules and internal voltage controls Fuel container Oxidizer container Reactant controls - H_2 (including plumbing, heat exchangers, etc.) Reactant controls - O_2 (includes plumbing, heat exchangers, etc.) Plumbing and valves Radiator, nonstructural (area) Electrical controls - coolant, reactant Pumps - coolant Structure Purge system Electrical coupling system (integral)
7.3 Power Source - Batteries	Battery (quantity)* Battery container and supports Voltage controls Recharge controls Structure Electrical coupling system (integral) Emergency *Identify system usage
7.4 Power Source - Solar Cell	Solar cell Intercell wiring Protective diodes (short regulators, etc.) Protective covers and filters Insulation and coatings Reflector Mounting panel or structure (solar paddle, etc.) Structure (including deployment devices) Orienting devices and controls Voltage controls Cooling system Electrical coupling system (integral) Photometric monitoring device

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	7. PRIME POWER SOURCE (Cont'd)
SECOND GENERATION	THIRD GENERATION
7.5 Power Source - Nuclear	Reactor or isotope source Radiation shield Primary systems (tubing, heaters, pumps) Power converter (rotating unit, boiler, structure, regulator, radiator) Heat rejection structure Electrical startup and controls Flight instrumentation (speed, etc.) Destruct system (less ordnance) Structure
7.6 Power Source - Gas Generator	Pressurization tank (force feed system) Pressurization controls (force feed system) Fuel containers Fuel controls Plumbing Valves and controls Decomposition chamber Speed regulator Gas bearing Gear box Turbine Heat exchanger Structure Purge system Voltage control Electrical coupling system (integral)
7.19 Contingency	
7.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	8. POWER CONVERSION AND DISTRIBUTION
SECOND GENERATION	THIRD GENERATION
8.1 Power Conversion - Electrical (AC)	Generator Transformers Inverters Phase adapters Frequency converters Voltage regulators Constant speed drive Supports
8.2 Power Conversion - Electrical (DC)	Generator Converters Voltage regulators Rectifiers Supports
8.3 Power Conversion - Hydraulic/Pneumatic	Motor pump or compressor Pump or compressor unit (unless integral) Reservoirs Accumulators Filters Pressure regulator Supports
8.4 Power Distribution - Electrical (AC)	Buses Isolator assembly Circuit breakers and switches Relays Electrical coupling Supports
8.5 Power Distribution - Electrical (DC)	Buses Isolation diode system Circuit breakers and switches Relays Electrical coupling Supports
8.6 Power Distribution - Hydraulic/Pneumatic	Valves Controls Plumbing Fluid Supports
8.7 Utility Provisions - Electrical	Pyrotechnic initiator Lights Signal devices
8.8 External Service Provisions (Type Power __)	
8.19 Contingency	
8.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	9. GUIDANCE AND NAVIGATION
SECOND GENERATION	THIRD GENERATION
9.1 Guidance Source - Inertial Reference Stellar Reference Planetary Reference Relative Reference	Programmer Accelerometers Gyros Telescope Sextant Horizon sensor Radar altimeter Rendezvous radar Rendezvous laser Optical sightline sensor Supports
9.2 Guidance Evaluation	Computer Computer keyboard Inertial platform Coupling unit Power and servo assembly Time base selector Gyro electronics Supports
9.3 Output	Electronics package Rate gyros Accelerometers Servo amplifiers Power supply Electrical coupling Supports
9.4 Spares	
9.19 Contingency	
9.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	10. INSTRUMENTATION
SECOND GENERATION	THIRD GENERATION
10.1 Sensors	Temperature Pressure Flow Volume Leak rate Acceleration Force Vibration Displacement Angular velocity Gas partial pressure Acoustic noise Strain Char and ablation Contamination Bio-medical Structure
10.2 Signal Conditioning	Analog commutator Analog-digital converter Digital commutator Programmer Calibrator Signal conditioning Data storage Structure
10.3 Signal Transmission	Antenna Waveguides Coax Transmitters Structure
10.4 Electrical Coupling	Junction box Cable harness Wire Coax Conduit Clips, clamps, etc. Structure
10.5 Support Items	
10.6 Spares	
10.19 Contingency	
10.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	11. COMMUNICATION
SECOND GENERATION	THIRD GENERATION
11.1 Intercommunication	Transceiver equipment Antenna system Structure
11.2 Near Earth Communication	VHF FM transmitter equipment VHF AM tranceiver equipment VHF antenna system Structure
11.3 Deep Space Communication	DSIF receiver equipment DSIF transmitter equipment DSIF power amplifier equipment DSIF antenna system Structure
11.4 TV Systems	Cameras Monitor Control equipment and components Structure
11.5 Tracking System	
11.6 Spares	Intercommunication Near earth Deep space Tracking system Television system
11.7 Electrical Coupling	Junction box Cable harness Wire Coax Conduit Clips, clamps, etc.
11.8 Racks & Supports	
11.19 Contingency	
11.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	12. ENVIRONMENTAL CONTROL
SECOND GENERATION	THIRD GENERATION
12.1 ECS - Equipment	Insulation Instrument temperature control system Electronic equipment temperature control system Compartment temperature control system Compartment pressurization system Engine cooling system (separate from engine) Fire control system Electrical coupling system Supports
12.2 ECS - Personnel	Insulation Temperature control system Atmosphere control system Pressurization control system Fire control system Electrical coupling system CO_2 and odor removal system Supports
12.3 Coolant System	Insulation Radiator Fluid boiler Plumbing and valves Blowers Heat exchangers Pumps Containers Supports
12.4 Multipurpose Equipment - Containers, Panels, etc.	Supports
12.19 Contingency	
12.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	13. ARMAMENT
SECOND GENERATION	THIRD GENERATION
13.1 Safety and Arming	Devices (less ordnance) Supports
13.2 Detonating	Devices (less ordnance) Supports
13.3 Warhead Provisions	Supports Checks Accessories
13.19 Contingency	
13.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	14. PERSONNEL PROVISIONS
SECOND GENERATION	THIRD GENERATION
14.1 Accommodations for Personnel	Seats Supports Restraints - seat Liners Shock absorbers Adjustment mechanisms Bunks Sleeping restraints
14.2 Fixed Life Support Equipment	Food containers Water containers Waste management Hygiene equipment Galley equipment
14.3 Cargo Handling	Rails Attachment fittings
14.4 Furnishings	Partitions Soundproofing Flooring (nonstructural) Lifts and hoist Stairs and ladders Platforms Catwalks
14.5 Emergency Equipment	Fire extinguisher (built-in) Fire detection Life raft provisions Life raft (built-in)
14.19 Contingency	
14.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	15. CREW STATION CONTROLS AND PANELS
SECOND GENERATION	THIRD GENERATION
15.1 Pedestal	Orientation, separation and ullage Propulsion Environment Navigation and guidance Prime power source Communication Instrumentation Range safety and abort
15.2 Control Stands	Orientation, separation and ullage Propulsion Environment Navigation and guidance Prime power source Communication Instrumentation Range safety and abort
15.3 Instrument Panels	Orientation, separation and ullage Propulsion Environment Navigation and guidance Prime power source Communication Instrumentation Range safety and abort
15.4 Crew Station Controls (Flight)	Control - roll - pitch - yaw Connecting members for above Supports
15.19 Contingency	
15.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	16. RANGE SAFETY AND ABORT
SECOND GENERATION	THIRD GENERATION
16.1 Sensors	
16.2 Signal Conditioning	
16.3 Signal Transmission	
16.4 Signal Evaluation	
16.5 Destruct System	Destruct receiver Range safety beacon Power supply Controls Supports
16.6 Electrical Coupling System	
16.7 Service Items	
16.8 Spares	
16.9 Debris, Re-entry	
16.19 Contingency	
16.20 Growth	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	17. PERSONNEL
SECOND GENERATION	THIRD GENERATION
17.1 Crew	Crew Constant wear garments Personnel comm. headset and belt pack Pressure suit
17.2 Personal Gear	Portable life support system Garments - protective Personal parachutes Hygienic equipment Privacy curtains
17.3 Life Support	Food Food containers (portable) Water Water drinking device Water containers (portable) Galley equipment (portable) Recreation equipment. Exercise equipment Medical and first aid equipment Survival kit (portable)
17.4 Crew Accessories	Maps and manuals Log book Maintenance tools Fire extinguisher (portable)

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	18. CARGO
SECOND GENERATION	THIRD GENERATION
18.1 Scientific Instruments	
18.2 Experiments	
18.3 Cargo	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	19. ORDNANCE
SECOND GENERATION	THIRD GENERATION
19.1 Propulsion Ordnance	Igniters Safe and arm Thrust termination Destruct
19.2 Orientation Controls, Separation and Ullage	Igniters (thrust systems) Separation
19.3 Prime Power Source	Destruct (nuclear systems)
19.4 Armament	Safe and arm Detonating Warhead
19.5 Range Safety and Abort	Destruct (vehicle)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	20. BALLAST
SECOND GENERATION	THIRD GENERATION

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	21. RESIDUAL PROPELLANT AND SERVICE ITEMS
SECOND GENERATION	THIRD GENERATION
21.1 Fuel Pressurizing Gas - As Container Residual	
21.2 Oxidizer Pressurizing Gas - As Container Residual	
21.3 Fuel Trapped - Main Engine	Tank Lines - between sump and engine pumps Engine - below pump inlet Sump Fill lines
21.4 Oxidizer Trapped - Main Engine	Tank Lines - between sump and engine pumps Engine - below pump lines Sump Fill lines
21.5 Fuel - Outage Main Engine	
21.6 Oxidizer - Outage Main Engine	
21.7 Fuel Trapped - Auxiliary Propulsion System	Tank Lines Engine
21.8 Oxidizer Trapped - Auxiliary Propulsion System	Tank Lines Engine
21.9 Fuel - Outage Auxiliary Propulsion System	
21.10 Oxidizer - Outage Auxiliary Propulsion System	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	21. RESIDUAL PROPELLANT AND SERVICE ITEMS (Cont'd)
SECOND GENERATION	THIRD GENERATION
21.11 Fuel Trapped - Electrical Power	Tank Lines Converter
21.12 Oxidizer Trapped - Electrical Power	Tank Lines Converter
21.13 Service Items Trapped - Including Non-expendables	Pressurizing gases (not in propellant tanks) Purge system gases Pneumatic system gases Air-bearing system gases Ejection system gases Hydraulic fluid Lubricants Fuel additives Environment control fluids Spatial attitude control propellants Propellants for landing or recovery

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	22. RESERVE PROPELLANT AND SERVICE ITEMS
SECOND GENERATION	THIRD GENERATION
22.1 Fuel Pressurizing Gas Reserves	
22.2 Oxidizer Pressurizing Gas Reserves	
22.3 Fuel - Main Engine Reserve	
22.4 Oxidizer - Main Engine Reserve	
22.5 Fuel Pressurizing Gas Reserves Auxiliary Propulsion Systems	
22.6 Oxidizer Pressurizing Gas Reserve - Auxiliary Propulsion Systems	
22.7 Fuel Reserves - Auxiliary Propulsion Systems	
22.8 Oxidizer Reserves - Auxiliary Propulsion Systems	
22.9 Fuel - Electrical Power Reserve	
22.10 Oxidizer - Electrical Power Reserve	
22.11 Fuel - Atmospheric Cruise	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	22. RESERVE PROPELLANT AND SERVICE ITEMS (Cont'd)
SECOND GENERATION	THIRD GENERATION
22.12 Oxidizer - Atmospheric Cruise	
22.13 Service Item Reserves	

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	23. INFLIGHT LOSSES (LIFTOFF TO SEPARATION)
SECOND GENERATION	THIRD GENERATION
23.1 Vented Pressuring Gas-Fuel System	
23.2 Vented Pressuring Gas-Oxidizer System	
23.3 Fuel - Auxiliary Propulsion System	Pressure relief losses Normal burning
23.4 Oxidizer - Auxiliary Propulsion System	Pressure relief losses Normal burning
23.5 Fuel - Electrical System	Pressure relief losses Normal burning
23.6 Oxidizer - Electrical System	Pressure relief losses Normal burning
23.7 Service Items	Pressurizing gases (not in propellant tanks) Purge system gases Pneumatic system gases Air-bearing system gases Ejection system gases Hydraulic fluid Lubricants Fuel additives Environmental control fluids Spatial attitude control propellants Frost Separation system propellants Ullage system propellants

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

FUNCTIONAL CODE

FIRST GENERATION	24. THRUST DECAY PROPELLANT
SECOND GENERATION	THIRD GENERATION
24.1 Fuel - Consumed During Thrust Decay	Container Sumps Lines Engines
24.2 Oxidizer - Consumed During Thrust Decay	Container Sumps Lines Engines
24.3 Solid Propellant - Consumed During Thrust Decay	
24.4 After-Cooling Propellants (Nuclear)	Container Sumps Lines Engines
24.5 Service Items	Pressurizing gases (not in propellant tanks) Purge system gases Pneumatic system gases Air-bearing system gases Ejection system gases Hydraulic fluid Lubricants Fuel additives Environmental control fluids Spatial attitude control propellants Frost Separation system propellants Ullage system propellants

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	25. FULL THRUST PROPELLANT
SECOND GENERATION	THIRD GENERATION
25. 1 Fuel	
25. 2 Oxidizer	
25. 3 Solid Propellants	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	26. THRUST BUILDUP PROPELLANT
SECOND GENERATION	THIRD GENERATION
26. 1 Fuel	Pressurizing gases Fuel
25. 2 Oxidizer	Pressurizing gases Oxidizer
26. 3 Solid Propellants	
26. 4 Auxiliary Propulsion System	
26. 5 Service Items	

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

FUNCTIONAL CODE

FIRST GENERATION	27. PRE-IGNITION LOSSES
SECOND GENERATION	THIRD GENERATION
27.1 Fuel	Pressurizing gases Fuel
27.2 Oxidizer	Pressurizing gases Oxidizer
27.3 Prime Power Source	
27.4 Service Items	Purge system gases Pneumatic system gases Air-bearing system gases Ejection system gases Hydraulic fluid Lubricants Fuel additives Environmental control fluids Spatial attitude control propellants Frost Emergency abort system propellants

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

APPENDIX C

MASS PROPERTIES DEPENDENT DESIGN INFORMATION

C.10.0 GENERAL DESIGN INFORMATION

This appendix lists the major dimensional factors and design criteria used in the development of mass properties estimating techniques and comparative studies (see par. 3.4.3.17.) The following criteria are general design items.

Ballistic coefficient (W/C_dA)
Actual drag coefficient
Mach number
Reynolds number
Lift-to-drag ratio
Type of guidance
Length
Maximum diameter
Width
Depth
Nose radius
Cross sectional area (maximum)
Wetted area
Pressurized body area
Window area
Drag and lift reference area
Total volume
Internal volume
Body volume pressurized
Cone half angle(s)
Stage weight-carrying limitations
Mission duration
Orbit altitudes
Total velocity increment required

C.20.0 FUNCTIONAL DESIGN INFORMATION

The following list of design information is intended as a guide for data submittal. Vehicle items are listed in accordance with the first-generation functional code (discussed in Appendix B).

1. Aerodynamic Surfaces (for each surface type)

Type
Material(s), types and properties
Aspect ratio
Fineness ratio
Safety factor
Span
Theoretical root chord - length
Theoretical root chord - thickness
Tip chord - length
Tip chord - thickness
Mean aerodynamic chord
Projected area
Wetted area
Sweepback at leading edge
Sweepback at 25% chord
Surface loading
Critical design conditions and resulting critical design loads

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:22Z
Check the source to verify that this is the current version before use.

2. Body Structure (for each major structural entity)

Type of structure (pressurized, monocoque, skin, stringer, etc.)
Material(s), types and properties
Safety factor
Length
Diameter
Skin thickness
Gross tank volumes
Tank design pressures
Maximum available volume for ballast
Design temperature
Maximum expected operating pressure
Center of pressure location and static margin
Critical design conditions and resulting critical design loads

3. Induced Environmental Protection (for each major environmental entity)

Material(s), types and properties
Manufacturing technique
Application method
Safety factor (value and definition)
Thicknesses
Gross area
Critical design conditions and resulting critical loads

4. Launch, Recovery and Docking

Design and operating parameters governing weight
Critical design conditions and resulting critical design loads

5. Main Propulsion

Number
Material(s), types and properties
Types of construction
Expansion ratio
Port-to-throat ratio
% of nozzle submerged
Shape of nozzle
Motor mass fraction (total propellant weight divided by total motor weight)
Specific impulse - sea level
Specific impulse - vacuum
Characteristic exhaust velocity
Throat action time
Wet action time
Thrust coefficient (C_f)
Type of thrust vector control
Total impulse
Chamber pressure (P_c) - average
Chamber pressure (P_c) - maximum
Critical design conditions and resulting critical design loads

5. (cont.) Liquid Engines

Number
Material(s), types and properties
Type of propellant feed
Type of construction
Type of cooling - thrust chamber

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

Type of cooling - nozzle extension
Number of starts
Throttling ratio
Expansion ratio
Characteristic exhaust velocity
Mixture ratio
Ignition type
Specific impulse - sea level
Specific impulse - vacuum
Thrust coefficient
Design burn time
Thrust - sea level
Thrust - vacuum
Throat area
Exit area
Chamber pressure (P_c) - average
Chamber pressure (P_c) - maximum
Net positive suction head
Propellant flow rate - total and components
Design propellant temperature range
Critical design conditions and resulting critical design loads

5. (cont.) Thrust Vector Control

Type
Roll control thrust
Roll angular acceleration requirement
Maximum axial thrust misalignment
Longitudinal angular acceleration requirement
Normal force produced
Critical design conditions and resulting critical design loads

5. (cont.) Other Propulsion Subsystems

Design and operating parameters governing weight
Critical design conditions and resulting critical design loads

6-20 Orientation Control, Separation and Ullage Through Ballast

Design and operating parameters governing weight
Critical design conditions and resulting critical loads

21-27 Propellants and Service Items

Solid Propellant(s)

Composition and properties
Loading ratio - propellant volume to case volume
Web fraction
Web thickness to propellant radius
Specific heat ratio
Storage/operating temperatures
Flame temperature
Characteristic exhaust velocity
Burning rate
Sliver percentage
Burning surface area

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the current version before use.

This page intentionally left blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

APPENDIX D

PARAMETERS AND INVENTORY OF FLUIDS AND PROPELLANTS LOADED

D.10.0 GENERAL

This appendix is intended as a guide for the presentation of data required for establishing the parameters and inventory of fluids and propellants loaded (as discussed in par. 3.4.3.23).

D.20.0 FLUID PROPELLANTS - MAIN ENGINE

D.20.1 Parameters of Maximum Loadable Weight. - The following parameters are for the total load limited by the engine inflight mixture ratio (MR) requirement on effective burning propellants.

a. Engine number(s)

b. MR of engine(s) - (indicate actual or specified).

1. Acceptance MR at standard conditions.

2. Average inflight MR (indicate whether inclusive of consumption by gas generators, vernier engines, etc. and include reference to the current MR production analysis).

c. Tankage volumes at loading and at total gross weight conditions for each propellant (indicate actual or specified).

1. Gross tankage (total cavity above letdown valve).

2. Required ullage volume (include reference to the current ullage analysis).

3. Net maximum loadable volume (c1 - c2).

4. Volume below each discrete tank level which is significant in establishing the loaded or terminal weight of propellant.

d. Propellant bulk temperature - each propellant (highest assumed before ignition).

e. Ullage pressures at loading and tanking secure.

f. Propellant bulk density - each propellant (lowest assumed before ignition).

g. Maximum loadable weight vs average inflight MR - each propellant and total propellant.

D.20.2 Inventory of Load. - The following inventory is for each propellant and total (lb).

a. Total nominal propellant loaded.

b. Propellant expended before liftoff (total and details of bleed, leakage, venting, start, holdown, etc.).

c. Total propellant aboard at liftoff (a - b).

d. Propellant expended during flight (total and details of bleed leakage, venting, effective burning, shutdown, etc. for each major flight stage before separation of stage being reported).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

e. Total propellant aboard at the end of each major flight stage.

f. Propellant aboard at separation of stage (total and details of trapped propellants and other system unusables), and reserve propellants (including propellant vapor retained).

D. 20. 3 Outage. - State any outage parameters, biases for outage if used, and indicate the mean and maximum probable outage. Account for these factors as used in the inventory. Include reference to the current outage prediction analysis.

D. 20. 4 Temperature and Density. Present the variations of loading bias, mean outage, maximum outage, average inflight mixture ratio, and propellant density with temperatures in the range around that assumed for loading at the scheduled launch period. Include reference to the current temperature prediction analysis.

D. 30. 0 OTHER FLUID LOADS

For all other fluid loads (nonpropellant), the following parameters and inventory apply.

D. 30. 1 Parameters of Maximum Loadable Weight. - State the parameters defining maximum loadable weight of each fluid, including, as applicable, gross volumes, limiting and net volumes, pressures, temperatures, gas constants or density.

D. 30. 2 Inventory of Load. - State the inventory of each fluid as loaded, and details of utilization, trapped and reserve quantities during the major flight stages and at separation of the stage.

D. 40. 0 SOLID PROPELLANTS LOADED

D. 40. 1 Parameters of Maximum Loadable Weight. - Identify the total case volume, the total volume occupied by the maximum loadable propellant, and the remaining volumes unoccupied by propellant. State all parameters which directly limit and establish the maximum loadable propellant.

D. 40. 2 Inventory of Propellant Loaded. - Inventory the propellant weights assigned to each principal phase of propellant utilization during flight and total same to the total propellant loaded in each solid motor segment or case.

D. 40. 3 Sliverage. - State values of the parameters used to establish the sliverage and indicate the mean and maximum sliverage. Include reference to the current sliverage prediction analysis.

D. 40. 4 Thrust/Time Curve. - Provide the current thrust/time curve which matches the weight of the propellant loaded.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:22Z
Check the source to verify that this is the currentversion before use.

APPENDIX E

MASS PROPERTIES VERIFICATION PLAN

E.10.0 GENERAL

This appendix is a guide for preparing the mass properties verification plan discussed in par. 3.4.3.25. The following items shall be included.

a. Analytical Verification Items.

1. Items to be verified by analysis.

2. Analytical substantiation that test is not required for these items.

b. Experimental Verification Items.

1. Items to be verified by experiment.

2. Analytical substantiation that test is required for these items.

3. Estimated costs of proposed tests.

E.20.0 TEST PLAN

For each item to be tested, or for each block of tests, list the following:

a. Measurement objectives, including measurement to be performed and overall accuracy required.

b. Description of the test setup, including the location, measurement system proposed, equipment, fixtures, pertinent dimensions and reference datum locations, and environmental control provisions.

c. Measurement procedures, number of successful runs required, the use of average readings from a number of independent measurements, and the requirements for complete unloading, unbalance or shutdown of equipment between measurement readings.

d. Instrumentation calibration procedure and schedule.

e. Statement of overall uncertainty of the measurement system, with supporting analysis of the system showing calibration and systematic error analysis. The measurement system analysis shall indicate possible sources of random error, their method of estimation, and their possible effect on the precision of measurement, and shall show the method of combining different sources of error to obtain a value for overall uncertainty of the measurement process. The analysis also shall indicate the relationship of the uncertainty with the required accuracy.

f. Schedule, including equipment availability dates and test start and completion dates.

g. Test reporting plan, including submittal schedules, for reports containing the following minimum information:

1. Explanation of deviations from test plan.

2. Summary of data.

3. Evaluation of results.

4. Final conclusion of results.

5. Recommendations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

This page intentionally left blank.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

APPENDIX F

CAPACITY AND LOADING INFORMATION FOR FLUIDS AND PROPELLANTS

F.10.0 GENERAL

This appendix lists the nominal information required to determine the parameters and procedures for pre- and post-flight evaluation of the loaded and unexpended fluid and propellant mass properties, as discussed in par. 3.4.3.24.

F.20.0 FLUID SYSTEMS

For fluid systems, provide the following information:

a. Sketches of liquid propellant and fluid system, on which are noted such items as tank and plumbing volumes, loading volumes, sensors, ports, net positive suction head (NPSH), vortex levels, "letdown" valve levels and other significant discrete level locations.

b. Derivation of total usable and trapped volumes; presentation of nominal liquid volumes vs station levels for the various temperature, pressure and acceleration conditions expected.

c. Synoptic derivation and discussion of minimum ullage volumes, with reference to the basic ullage analysis documents.

d. Synoptic derivation and discussion of techniques for predicting the mean and maximum propellant outage, with reference to the basic outage prediction analysis.

e. Synoptic description and discussion of anticipated propellant temperature and rise rates at the launch site for each calender month, with reference to the basic temperature prediction program.

f. Synoptic descriptions of loading methods and procedures for control of the fluid weight as loaded, with reference to the basic methods and procedures documents.

g. Presentation of gas constants (R), propellant and liquid densities vs temperature, pressure and other environments expected.

h. Synoptic descriptions of test and flight instrumentation and procedures to determine the initial loadings and unexpended propellant and fluid weights from flight and static tests.

F.30.0 SOLID SYSTEMS

For solid systems, provide the following information:

a. Sketches of propellant system in plan, and significant cross-sections showing stations and dimensions for:

1. Grain configuration at initial loading and ignition.

2. Significant burntime grain configurations.

3. Typical terminal configurations of sliverage at end of web action time and end of action time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

4. Liner, insulation and bonding.

5. Igniter and thrust termination systems.

b. Volumetric data for:

1. Total case capacity.

2. Propellant, insulation and liner for conditions (a-1), (a-2), and (a-3) above.

c. Densities of propellants, insulation, liner and bonding materials.

d. Synoptic description of loading methods and procedures for control of the solid weight as loaded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE AIR FORCE

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

SD/ALM
P. O. Box 92960
Worldway Postal Center
Los Angles, CA 90009

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T12:22Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:22Z
Check the source to verify that this is the current version before use.